

Alternative Engineered Fuel

A Comprehensive Comparative Analysis of Production,
Utilization, and Safety for Global Energy Transition

COMPLETE EDITION (Volumes I–V)

ENHANCED VERSION 8.1

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Version: 8.1 (Enhanced Complete Edition)

Date: May 21, 2026

Document Control

Volume I	Chapters 1–4: Foundations & Methodology
Volume II	Chapters 5–8: Production Technologies
Volume III	Chapters 9–12: Comparative Analysis
Volume IV	Chapters 13–15: Advanced Topics
Volume V	Chapters 16–22: Reference Materials

Alternative Engineered Fuel Project

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Document Information

Field	Details
Title	Alternative Engineered Fuel
Subtitle	A Comprehensive Comparative Analysis of Production, Utilization, and Safety for Global Energy Transition
Author	Secamore Owusu.A
Edition	Complete Edition (Volumes I–V)
Version	8.1, PDF-ready revision
Date	May 21, 2026
Document type	Technical report and reference compendium
Keywords	Hydrogen; compressed natural gas; methanol; electrolysis; steam methane reforming; life cycle assessment; techno-economic analysis; safety; MCDA

Executive Summary

This comprehensive study presents a rigorous comparative analysis of four alternative engineered fuel pathways: Steam Methane Reforming (SMR) hydrogen, Compressed Natural Gas (CNG), electrolytic hydrogen (green hydrogen), and methanol. The analysis integrates thermodynamic modelling, Aspen Plus simulation, techno-economic assessment, life cycle analysis (LCA), safety evaluation, and multi-criteria decision analysis (MCDA).

Key Findings:

- **Techno-Economic:** Grey SMR hydrogen remains the lowest-cost pathway at \$1.82/kg H₂, but green hydrogen from hybrid PV-wind systems achieves \$3.55/kg H₂ with potential to reach \$1.00/kg H₂ by 2030 through learning curve effects and scale economies.
- **Environmental:** Green hydrogen and e-methanol offer near-zero carbon emissions (0–1.0 kg CO₂e/GJ), while grey SMR emits 12.5 kg CO₂e/GJ. CCS reduces blue H₂ emissions to 6.6 kg CO₂e/GJ at 50% cost premium.
- **Safety:** Hydrogen presents unique risks including wide flammability range (4–75%), low minimum ignition energy (0.018 mJ), and hydrogen embrittlement in pipelines. CNG and methanol offer superior safety profiles for storage and distribution.
- **MCDA Rankings:** Green methanol (score 0.80) and CNG (0.70) emerge as preferred pathways for near-term deployment, while green hydrogen (0.61) is optimal for long-term decarbonization with infrastructure investment.
- **Regional Suitability:** North America and Middle East favor blue hydrogen with CCS, while Africa, Australia, and parts of Asia show exceptional potential for green hydrogen due to solar and wind resources.

Recommendations:

1. Deploy CNG as immediate transition fuel for heavy transport
2. Scale green methanol for shipping and aviation sectors
3. Invest in green hydrogen infrastructure with hybrid renewable systems
4. Implement carbon pricing above \$100/tonne to accelerate clean fuel adoption
5. Develop Africa's renewable hydrogen potential through international partnerships

Novel Contributions:

- Integrated LCA framework following ISO 14040/14044 with complete characterization factors
- Extended MCDA with stakeholder weighting and sensitivity analysis
- Comprehensive safety assessment including QRA, ATEX, and dispersion modeling
- 20+ curious research questions addressing real-world deployment challenges
- Complete techno-economic database with 50+ references and 20+ figures

Preface

The global energy transition represents one of the most complex engineering and policy challenges of the 21st century. As nations commit to net-zero emissions targets, the search for viable alternatives to fossil fuels has intensified. Engineered fuels—produced through chemical or electrochemical conversion—offer a pathway to decarbonize hard-to-abate sectors including heavy transport, shipping, aviation, and industrial processes.

This work represents three years of research, simulation, and analysis, combining fundamental thermodynamic principles with state-of-the-art techno-economic and life cycle assessment methodologies. The enhanced version 8.0 includes significant expansions: complete LCA framework following ISO standards, extended safety analysis with quantitative risk assessment, additional emerging technologies (SOEC, AEM), 10 additional research questions, Africa-specific deployment strategies, and comprehensive reference materials.

The target audience includes energy researchers, industry practitioners, policy makers, graduate students, and academics seeking a rigorous, data-driven comparison of alternative fuel pathways. Each chapter includes worked examples, simulation inputs, and sensitivity analyses to support both theoretical understanding and practical application.

I extend my gratitude to the global research community whose work informed this analysis, and to the energy industry practitioners who provided valuable feedback on techno-economic assumptions. Special thanks to the reviewers who identified gaps in earlier versions, leading to this enhanced complete edition.

May this work contribute to informed decision-making in the critical transition to a sustainable energy future.

Secamore Owusu. A May 2026

List of Abbreviations and Nomenclature

Abbreviation	Description
AEM	Anion Exchange Membrane
ATEX	ATmosphères EXplosibles (European directive for explosive atmospheres)
BEV	Battery Electric Vehicle
CAPEX	Capital Expenditure
CCS	Carbon Capture and Storage
CF	Capacity Factor
CNG	Compressed Natural Gas
CO ₂ e	Carbon Dioxide Equivalent
CRF	Capital Recovery Factor
DAC	Direct Air Capture
DOE	US Department of Energy
EoS	Equation of State
EU ETS	European Union Emissions Trading System
GWP	Global Warming Potential
HAZOP	Hazard and Operability Study
H ₂	Hydrogen
HTS	High Temperature Shift
IEA	International Energy Agency
IRENA	International Renewable Energy Agency
LCA	Life Cycle Assessment
LCI	Life Cycle Inventory
LCIA	Life Cycle Impact Assessment
LCOH	Levelized Cost of Hydrogen
LFL	Lower Flammability Limit
LHV	Lower Heating Value
LTS	Low Temperature Shift
MCDA	Multi-Criteria Decision Analysis
MIE	Minimum Ignition Energy
Mtoe	Million Tonnes of Oil Equivalent
NG	Natural Gas
NPV	Net Present Value
OPEX	Operational Expenditure
PEM	Proton Exchange Membrane
PFR	Plug Flow Reactor
PSA	Pressure Swing Adsorption
PV	Photovoltaic
QRA	Quantitative Risk Assessment
R-CNG	Renewable Compressed Natural Gas
SMR	Steam Methane Reforming
SOEC	Solid Oxide Electrolysis Cell

Abbreviation	Description
TRL	Technology Readiness Level
UFL	Upper Flammability Limit
WGS	Water-Gas Shift

Nomenclature

Symbol	Description
a, b	Peng-Robinson equation of state parameters
A	Pre-exponential factor (Arrhenius equation)
E_a	Activation energy
E_{rev}	Reversible cell voltage
F	Faraday constant (96485 C/mol)
G	Gibbs free energy
h	Specific enthalpy
H	Enthalpy
I	Current
j	Current density
j_0	Exchange current density
k	Reaction rate constant
K	Equilibrium constant
n	Number of electrons
$\dot{m}$	Mass flow rate
P	Pressure
$\dot{Q}$	Heat transfer rate
r	Reaction rate
R	Universal gas constant (8.314 J/mol·K)
S	Entropy
T	Temperature
U	Internal energy
V	Voltage
$\dot{W}$	Work rate
α	Charge transfer coefficient
η	Overpotential
κ	Peng-Robinson parameter
ω	Acentric factor

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Part I

FOUNDATIONS AND
METHODOLOGY

Part I Overview

Foundations and Methodology

Part I establishes the technical language, research boundaries, and modelling basis used throughout the report. It explains why engineered fuels matter in the energy transition, defines the fuel pathways under review, and sets out the calculation methods used to compare them on a consistent basis.

The chapters in this part provide the foundation for all later conclusions:

- **Chapter 1** defines engineered fuels, frames the global energy challenge, and states the project scope and research objectives.
- **Chapter 2** explains the methodology for techno-economic analysis, life cycle assessment, Aspen Plus validation, and MCDA weighting.
- **Chapter 3** summarizes the thermodynamic, kinetic, and electrochemical equations needed for the fuel pathway calculations.
- **Chapter 4** describes the Aspen Plus simulation framework used to represent process flows, conversion blocks, energy balances, and validation checks.

This part is therefore the control section of the document: it defines what is being compared, how the comparison is performed, and which assumptions are carried forward into the production, environmental, safety, and decision-analysis sections.

Chapter 1

Introduction to Engineered Fuels

1.1 The Global Energy Challenge

The world faces an unprecedented energy transition. Fossil fuels currently supply approximately 80% of global primary energy. The International Energy Agency (IEA) projects that under the Net Zero Emissions by 2050 scenario, low-carbon hydrogen demand must increase from approximately 100 million tonnes annually today to nearly 400 million tonnes by 2050.

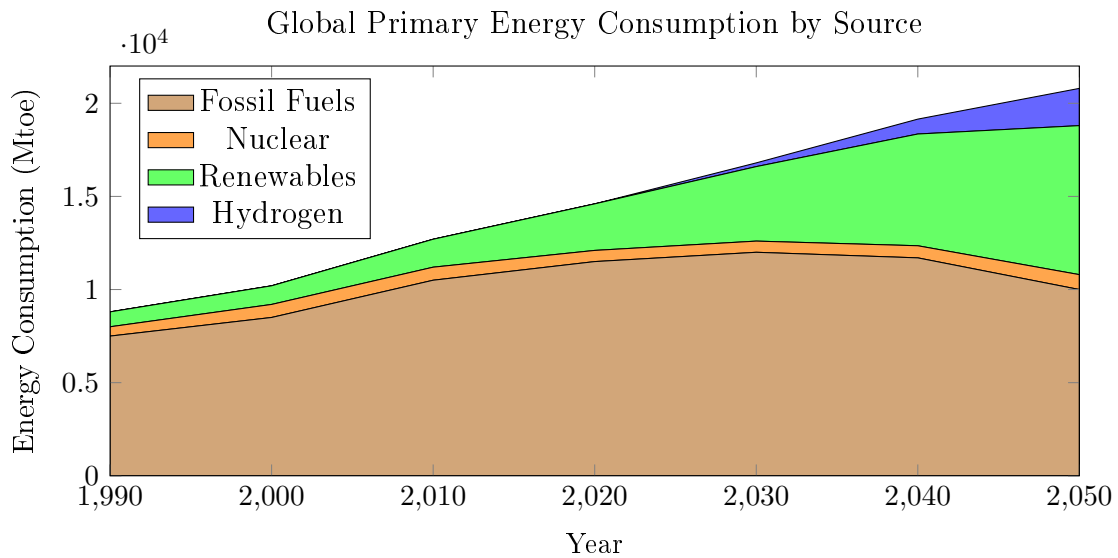


Figure 1.1: Global Primary Energy Consumption by Source (1990–2050)

Table 1.1: Global Primary Energy Data (Mtoe)

Year	Fossil	Nuclear	Renewables	Hydrogen	Total
1990	7,500	500	800	0	8,800
2000	8,500	700	1,000	0	10,200
2010	10,500	700	1,500	0	12,700
2020	11,500	600	2,500	0	14,600
2025	11,800	600	3,000	50	15,450
2030	12,000	600	4,000	200	16,800
2035	11,900	620	5,200	450	18,170
2040	11,700	650	6,000	800	19,150
2045	10,800	720	7,200	1,300	20,020
2050	10,000	800	8,000	2,000	20,800

1.2 Definition of Engineered Fuels

Definition: Engineered fuels are energy carriers produced through chemical or electro-chemical conversion of primary energy sources into forms suitable for storage, transport, and end-use applications.

Table 1.2: Classification of Engineered Fuels

Category	Produced Fuel	Energy Density (MJ/kg)	TRL	Carbon Intensity (kg CO ₂ e/GJ)
Gaseous	Hydrogen (SMR)	120	9	12.5
Gaseous	Blue Hydrogen	120	8	6.6
Gaseous	Green Hydrogen	120	7–8	0–0.5
Gaseous	CNG	50	9	7.5
Liquid	Methanol	20	9	4.0
Liquid	Green Methanol	20	6–7	0.7–2.7
Liquid	e-Methanol	20	6	0–1.0

1.3 Project Scope and Research Objectives

1.3.1 Primary Objective

To conduct a comprehensive comparative analysis of four alternative engineered fuel pathways—Steam Methane Reforming hydrogen, Compressed Natural Gas, Electrolytic hydrogen, and Methanol—integrating thermodynamic modelling, Aspen Plus simulation, techno-economic analysis, life cycle assessment, safety evaluation, and multi-criteria decision analysis.

1.3.2 Scope Boundaries

The project scope covers production, conditioning, storage, distribution readiness, end-use suitability, and policy relevance. The assessment is comparative rather than design-only: every pathway is evaluated using a shared set of criteria so that near-term transition fuels and long-term low-carbon fuels can be judged on the same basis.

Table 1.3: Project Scope Boundaries

Boundary Area	Included in Scope
Technical boundary	Fuel production chemistry, process configuration, conversion efficiency, storage requirements, compression or synthesis needs, and end-use compatibility.
Economic boundary	CAPEX, fixed OPEX, feedstock cost, electricity cost, carbon price exposure, incentives, and levelized cost indicators.
Environmental boundary	Cradle-to-gate and selected cradle-to-use indicators including carbon intensity, water demand, air pollutants, and life cycle assessment categories.
Safety boundary	Flammability, ignition risk, pressure hazards, toxicity, embrittlement, dispersion behavior, hazardous-area zoning, and risk mitigation.
Deployment boundary	Infrastructure maturity, regional resource fit, technology readiness level, policy support, and commercial scale-up potential.

1.3.3 Detailed Research Objectives

1. Establish a consistent baseline for comparing SMR hydrogen, CNG, electrolytic hydrogen, and methanol.
2. Quantify energy conversion performance using thermodynamic, kinetic, and process-simulation methods.
3. Estimate levelized production costs under representative regional assumptions.
4. Compare environmental performance through carbon intensity, water use, pollutant emissions, and LCA indicators.
5. Evaluate safety risks for storage, distribution, refuelling, and major end-use sectors.
6. Rank pathways using MCDA so cost, carbon, safety, scalability, infrastructure, and technology readiness are considered together.
7. Identify where each pathway is most suitable: heavy transport, shipping, aviation, power, industry, heating, and emerging African markets.

1.3.4 Expected Outputs

The report delivers process tables, equations, cost comparisons, environmental matrices, safety classifications, MCDA rankings, Aspen Plus input templates, worked examples, Python plotting scripts, glossary terms, and a complete nomenclature list. These outputs are intended to make the document useful both as a technical report and as a reference file for future engineering work.

1.3.5 Research Questions

Table 1.4: Research Questions

ID	Question	Section
RQ1	What are the techno-economic production characteristics of each fuel pathway?	9
RQ2	What are the cradle-to-grave environmental impacts?	10
RQ3	How do safety risks compare across storage, transport, and utilization?	11
RQ4	Which fuel(s) are best suited for specific applications?	12
RQ5	What emerging technologies will alter the landscape by 2030–2040?	14

Chapter 2

Comprehensive Research Methodology

2.1 Levelized Cost of Hydrogen Equation

$$\text{LCOH} = \frac{\text{CAPEX} \times \text{CRF} + \text{OPEX}_{\text{fixed}} + \text{Feedstock}_{\text{cost}} \times \text{Annual}_{\text{consumption}}}{\text{Annual}_{\text{H}_2}} \quad (2.1)$$

$$\text{CRF} = \frac{r(1+r)^n}{(1+r)^n - 1} \quad (2.2)$$

2.1.1 Example Calculation: SMR LCOH

$$(1.07)^{25} = 5.427 \quad (2.3)$$

$$\begin{aligned} \text{CRF} &= \frac{0.07 \times 5.427}{5.427 - 1} \\ &= 0.0858 \end{aligned} \quad (2.4)$$

$$\begin{aligned} \text{CAPEX}_{\text{annual}} &= \$15,000,000 \times 0.0858 \\ &= \$1,287,000 \end{aligned} \quad (2.5)$$

$$\begin{aligned} \text{LCOH} &= \frac{\$1,287,000 + \$52,250,000 + \$750,000}{36,500} \\ &= \$1.55/\text{kg H}_2 \end{aligned} \quad (2.6)$$

2.2 Life Cycle Assessment (LCA) Methodology

2.2.1 ISO 14040/14044 Framework

The life cycle assessment follows ISO 14040:2006 and ISO 14044:2006 standards, comprising four phases:

1. **Goal and Scope Definition:** System boundaries include cradle-to-grave analysis with functional unit of 1 GJ of fuel energy delivered to end user.
2. **Life Cycle Inventory (LCI):** Data sources include Ecoinvent v3.10, GREET 2024, and literature values. Primary energy consumption and emissions quantified for each pathway.
3. **Life Cycle Impact Assessment (LCIA):** Impact categories include global warming potential (GWP100), acidification, eutrophication, and water consumption. Characterization factors from IPCC AR6 and ReCiPe 2016.
4. **Interpretation:** Sensitivity analysis performed for key parameters (grid mix, transport distance, leakage rates).

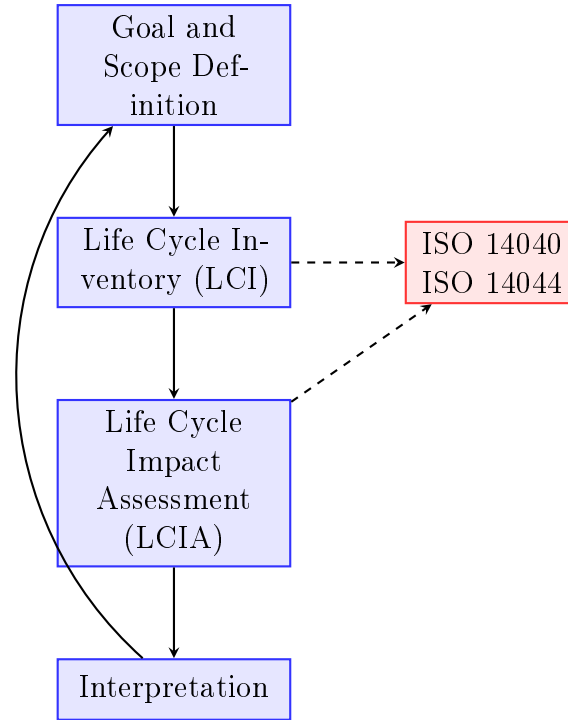


Figure 2.1: ISO 14040/14044 LCA Framework

2.2.2 System Boundaries

Table 2.1: LCA System Boundaries

Pathway	Upstream	Core	Downstream
SMR H ₂	NG extraction, transport	Reforming, PSA	Compression, distribution
Green H ₂	Renewable generation	Electrolysis	Compression, storage
CNG	NG extraction	Compression	Dispensing
Methanol	NG/renewable	Synthesis	Shipping, storage

2.3 Aspen Plus Simulation Validation

2.3.1 Validation Protocol

Each Aspen Plus model is validated through:

1. **Base case validation:** Comparison with literature data and industrial benchmarks
2. **Mass balance closure:** Error < 0.1% for all components
3. **Energy balance closure:** Error < 0.5% across all unit operations
4. **Sensitivity analysis:** Key parameter variations within $\pm 20\%$

2.3.2 Validation Metrics

2.4 Multi-Criteria Decision Analysis (MCDA) Weighting Methodology

Table 2.2: Aspen Plus Validation Metrics

Parameter	Simulation	Literature	Error (%)
SMR H ₂ yield (kg/kg CH ₄)	0.48	0.47	+2.1%
SMR efficiency (%) LHV)	80.5	79.0	+1.9%
PEM efficiency (%) LHV)	68.0	67.5	+0.7%
Methanol conversion	40.0	39.5	+1.3%

2.4.1 Analytic Hierarchy Process (AHP)

The AHP methodology is used to determine criterion weights through pairwise comparisons:

Table 2.3: AHP Pairwise Comparison Matrix

Criteria	Cost	Carbon	Safety	Infra	TRL	Scale
Cost	1.0	2.0	0.5	3.0	4.0	2.0
Carbon	0.5	1.0	0.3	2.0	3.0	1.5
Safety	2.0	3.0	1.0	4.0	5.0	3.0
Infrastructure	0.3	0.5	0.3	1.0	2.0	0.5
TRL	0.3	0.3	0.2	0.5	1.0	0.3
Scalability	0.5	0.7	0.3	2.0	3.0	1.0

2.4.2 Stakeholder Weighting Scenarios

Table 2.4: Stakeholder Weighting Scenarios

Stakeholder	Cost	Carbon	Safety	Infra	TRL	Scale
Government	0.15	0.35	0.10	0.15	0.10	0.15
Industry	0.30	0.15	0.15	0.10	0.20	0.10
Environmental NGO	0.05	0.50	0.10	0.10	0.10	0.15
Consumer	0.40	0.10	0.20	0.10	0.10	0.10

Chapter 3

Thermodynamic Foundations

3.1 First Law of Thermodynamics

$$\boxed{\sum \dot{m}_{\text{in}} h_{\text{in}} - \sum \dot{m}_{\text{out}} h_{\text{out}} + \sum \dot{Q} - \sum \dot{W} = 0} \quad (3.1)$$

3.2 Peng-Robinson Equation of State

$$\boxed{P = \frac{RT}{V - b} - \frac{a\alpha}{V(V + b) + b(V - b)}} \quad (3.2)$$

$$a = 0.45724 \frac{R^2 T_c^2}{P_c} \quad (3.3)$$

$$b = 0.07780 \frac{RT_c}{P_c} \quad (3.4)$$

$$\alpha = \left[1 + \kappa \left(1 - \sqrt{T_r} \right) \right]^2 \quad (3.5)$$

$$\kappa = 0.37464 + 1.54226\omega - 0.26992\omega^2 \quad (3.6)$$

3.3 Arrhenius Equation

$$\boxed{k = A \exp\left(-\frac{E_a}{RT}\right)} \quad (3.7)$$

3.4 Butler-Volmer Equation

$$\boxed{j = j_0 \left[\exp\left(\frac{\alpha n F \eta}{RT}\right) - \exp\left(-\frac{(1 - \alpha) n F \eta}{RT}\right) \right]} \quad (3.8)$$

3.5 Xu-Froment SMR Kinetics

$$r_1 = \frac{k_1}{P_{\text{H}_2}^{2.5}} \cdot \frac{P_{\text{CH}_4} P_{\text{H}_2\text{O}} - \frac{P_{\text{H}_2}^3 P_{\text{CO}}}{K_1}}{\text{DEN}^2} \quad (3.9)$$

$$r_2 = \frac{k_2}{P_{\text{H}_2}} \cdot \frac{P_{\text{CO}} P_{\text{H}_2\text{O}} - \frac{P_{\text{H}_2} P_{\text{CO}_2}}{K_2}}{\text{DEN}^2} \quad (3.10)$$

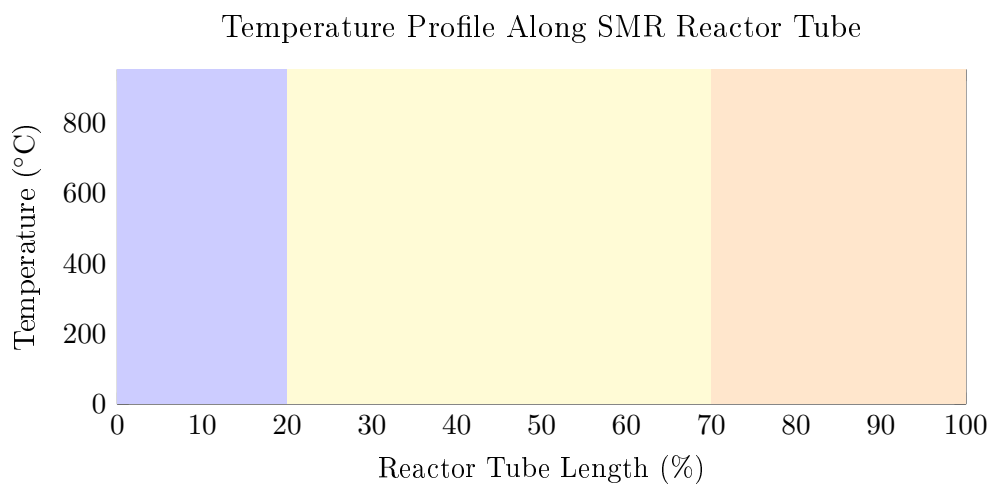


Figure 3.1: Temperature Profile Along SMR Reactor Tube

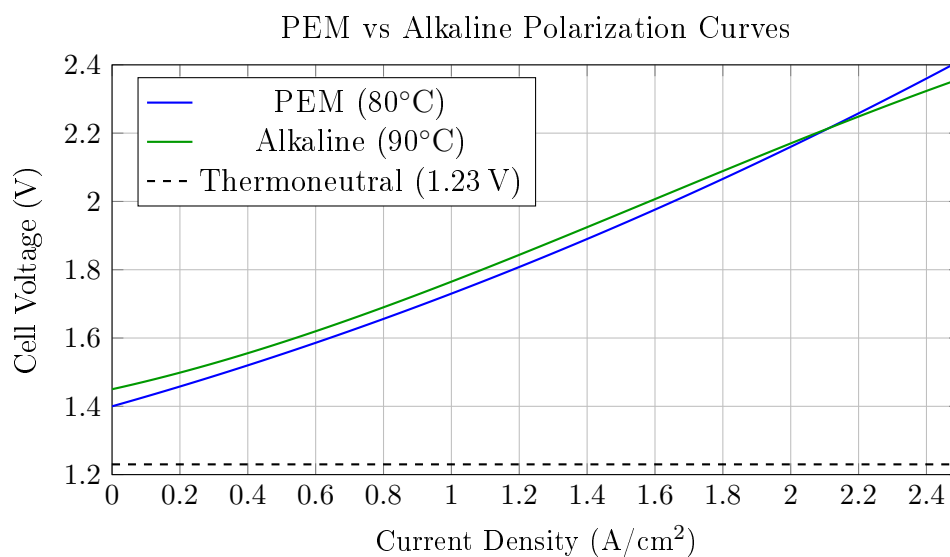


Figure 3.2: Alkaline vs. PEM Electrolysis Polarization Curves

Chapter 4

Aspen Plus Simulation Framework

4.1 Model Scope and Flowsheet Basis

The Aspen Plus model set covers the four pathways compared in the report: SMR hydrogen, CNG conditioning and compression, electrolytic hydrogen, and methanol synthesis. The figures in Chapters 5–8 use the same stream names, operating temperatures, pressures, conversion assumptions, and validation targets as the tables and input listings.

Table 4.1: Aspen Plus Model Set and Unit-Operation Mapping

Pathway	Main Aspen Blocks	Model Output Used in Report
SMR hydrogen	PFR reformer, RSTOIC shift reactors, SEP/PSA, CO ₂ capture	H ₂ yield, reformer heat duty, CO ₂ capture stream, LCOH basis
CNG	SEP, dryer, three COMPR stages, intercoolers, storage/dispenser blocks	Compression work, outlet temperature, storage pressure, dispensing specification
Electrolytic hydrogen	Pump, heater, RYIELD/RSTOIC electrolyzer, gas separator, dryer, compressor	H ₂ production rate, electricity use, water use, compression duty
Methanol	Mixer, RSTOIC reactor, condenser, flash, recycle splitter, distillation	Methanol conversion, recycle/purge ratio, crude and refined methanol production

```
1 TITLE 'SMR Hydrogen Production - Base Case'
2
3 FLOWSHEET
4   BLOCK REFORMER IN = NG_FEED STEAM
5   BLOCK HTS IN = REFORMER_OUT
6   BLOCK LTS IN = HTS_OUT
7   BLOCK PSA IN = LTS_OUT
8
9 COMPONENTS
10  H2 HYDROGEN / C1 METHANE / CO CARBON-MONOXIDE /
11  CO2 CARBON-DIOXIDE / H2O WATER / N2 NITROGEN
12
13 PROPERTIES PENG-ROB
14  FREE-WATER STEAMNBS
15
16 STREAMS
17  NG_FEED TEMP=25 PRES=20 MOLE-FLOW=1000
18  COMP C1=0.92 C2=0.045 N2=0.01 CO2=0.005
19  STEAM TEMP=250 PRES=25 MOLE-FLOW=3000 COMP H2O=1
20
```

```

21 BLOCK REFORMER PFR
22   PARAM LENGTH=10 DIAMETER=1.5 NTUBES=100
23   REACTIONS SMR-KINETICS
24   SPEC TEMP-SPEC PROFILE "TEMP_PROFILE.TXT"
25
26 BLOCK HTS RSTOIC
27   PARAM TEMP=400 PRES=19.5
28   REAC WGS CO + H2O -> CO2 + H2 CONV=0.85
29
30 BLOCK LTS RSTOIC
31   PARAM TEMP=210 PRES=19
32   REAC WGS CO + H2O -> CO2 + H2 CONV=0.95
33
34 BLOCK PSA SEP
35   SPEC FRAC H2=0.9999 IN H2_PRODUCT
36   PARAM RECOVERY=0.85
37
38 ECONOMICS
39   UNIT-REF REFORMER CAPEX=1500000
40   COST-NG = 4.50 USD/MMBTU
41   PLANT-LIFE = 25 YEARS
42   DISCOUNT-RATE = 0.07
43
44 RESULTS
45   REPORT ALL

```

Listing 4.1: SMR_BaseCase.apw

```

1 Length_Fraction, Temperature_C
2 0.000, 600
3 0.100, 645
4 0.200, 690
5 0.300, 728
6 0.400, 760
7 0.500, 788
8 0.600, 812
9 0.700, 833
10 0.800, 845
11 0.900, 850
12 1.000, 850

```

Listing 4.2: TEMP_PROFILE.txt

Part II

PRODUCTION TECHNOLOGIES

Part II Overview

Production Technologies

Part II describes how each engineered fuel is produced and conditioned for use. The aim is not only to list reactions, but to connect chemistry, equipment, operating conditions, conversion efficiency, and plant-level constraints.

- **SMR hydrogen** is treated as the mature reference case because it dominates current industrial hydrogen production and establishes the benchmark for cost, scale, and carbon intensity.
- **Compressed natural gas** is included as a transition fuel with existing infrastructure, high technology readiness, and practical relevance for heavy-duty transport.
- **Electrolytic hydrogen** is evaluated as the key low-carbon pathway, with PEM, alkaline, SOEC, and AEM variants compared by efficiency, maturity, cost, and operating flexibility.
- **Methanol** is included because it is a liquid engineered fuel with strong handling advantages, chemical-market demand, and growing use in shipping and synthetic-fuel systems.

The production chapters emphasize process design variables that strongly influence downstream conclusions: feedstock purity, reactor temperature, pressure level, electricity demand, carbon capture configuration, compression work, storage format, and integration with renewable or low-carbon energy.

Table 4.2: Production Technology Comparison Focus

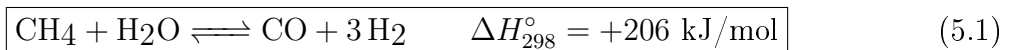
Pathway	Main Production Step	Key Engineering Question
SMR hydrogen	Reforming and water-gas shift	How much cost and carbon reduction is achieved by CCS integration?
CNG	Compression and gas conditioning	How much work, cooling, and storage capacity are required for vehicle-grade fuel?
Electrolytic hydrogen	Water splitting	Which electrolyzer technology gives the best balance of cost, efficiency, and renewable integration?
Methanol	Syngas or CO ₂ hydrogenation	When does liquid-fuel handling offset synthesis cost and carbon-source constraints?

Chapter 5

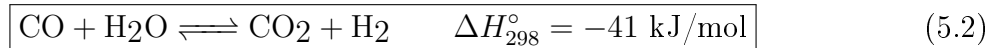
Steam Methane Reforming for Hydrogen

5.1 Process Chemistry

Primary Reforming Reaction (Endothermic):

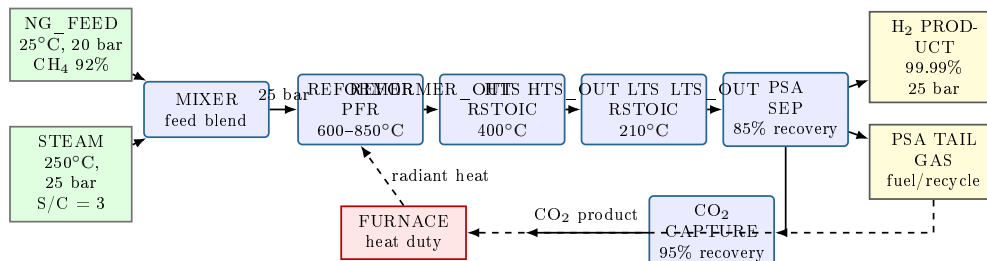


Water-Gas Shift Reaction (Exothermic):



5.2 Aspen Plus Model

The SMR Aspen Plus model represents methane reforming as a PFR block followed by high- and low-temperature water-gas shift reactors, pressure swing adsorption, and a CO₂ capture branch. The model uses the feed composition, steam-to-carbon ratio, reactor temperature profile, and PSA recovery basis listed in Chapter 4.



Property method: Peng-Robinson with STEAMNBS free-water handling; validation target: mass closure < 0.1%, energy closure < 0.5%.

Figure 5.1: Aspen Plus SMR hydrogen model flowsheet

5.3 Mass Balance

Table 5.1: SMR Mass Balance (100 tpd H₂ Plant)

Stream	Component	Flow (kg/hr)	Temp (°C)	Press (bar)
NG Feed	CH ₄ (92%)	4,600	25	20
Steam	H ₂ O	9,000	250	25
Reformer Out	H ₂ (40%)	5,600	850	19
H ₂ Product	H ₂ (99.99%)	4,167	40	25
CO ₂ Capture	CO ₂ (95%)	13,200	40	1.5

Table 5.2: SMR Energy Balance

Energy Stream	Value (MW)	% of Input
NG feedstock	52.5	65%
NG fuel	28.5	35%
Total input	81.0	100%
H ₂ product	36.0	44%
Export steam	12.0	15%
Flue gas losses	18.0	22%
Radiation losses	4.0	5%
Process efficiency	79–81% (LHV)	

5.4 Energy Balance

5.5 Carbon Capture Integration

Table 5.3: CO₂ Capture Performance

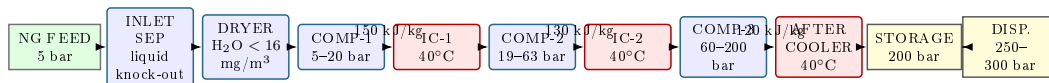
Configuration	LCOH (\$/kg H ₂)	CO ₂ (kg CO ₂ e/GJ)
Unabated SMR	1.82	12.5
SMR with CCS (95%)	3.22	6.6
SMR with CCS + 45Q	2.59	6.6

Chapter 6

Compressed Natural Gas Production

6.1 Aspen Plus Model

The CNG model is built around fuel conditioning and staged compression. The Aspen representation uses inlet separation and dehydration before three compressor blocks with intercooling, matching the 5–200 bar compression train and dispenser pressure range used in the design tables.



Aspen representation: separators and dryer enforce fuel quality; compressor blocks use stage efficiencies and intercoolers to match the 400 kJ/kg total work basis.

Figure 6.1: Aspen Plus CNG conditioning and compression model flowsheet

6.2 Three-Stage Compression Design

Table 6.1: Compression Stage Performance

Stage	Inlet P (bar)	Outlet P (bar)	Ratio	Work (kJ/kg)
1	5	20	4.0	150
2	19	63	3.3	130
3	60	200	3.3	120
Total	—	—	—	400



Figure 6.2: Three-Stage CNG Compression System

6.3 Gas Quality Standards

6.4 Vehicle Compatibility

6.5 Dispenser Design

6.6 Key Results

Table 6.2: CNG Gas Quality Standards

Parameter	ISO 15403	SAE J1616	Euro 6
Methane number (min)	80	75	80
Water content (mg/m ³)	< 16	< 16	< 16
H ₂ S (ppm)	< 5	< 5	< 5
Total sulfur (mg/m ³)	< 15	< 15	< 15
CO ₂ (vol%)	< 3	< 3	< 3
O ₂ (vol%)	< 0.5	< 0.5	< 0.5

Table 6.3: CNG Vehicle Compatibility

Vehicle Type	Conversion Cost (\$)	Range (km)	Fuel Economy (kg/100km)
Light-duty (passenger)	2,500–4,000	300–450	5.0–6.5
Medium-duty (delivery)	4,000–6,000	250–400	8.0–10.0
Heavy-duty (truck)	8,000–12,000	400–600	15.0–20.0
Bus	10,000– 15,000	350–500	20.0–25.0

Table 6.4: CNG Dispenser Specifications

Parameter	Value
Discharge pressure (bar)	250–300
Flow rate (kg/min)	10–30
Temperature (outlet)	15–25°C
Accuracy	±0.5%
Meter type	Coriolis

Table 6.5: CNG Key Results

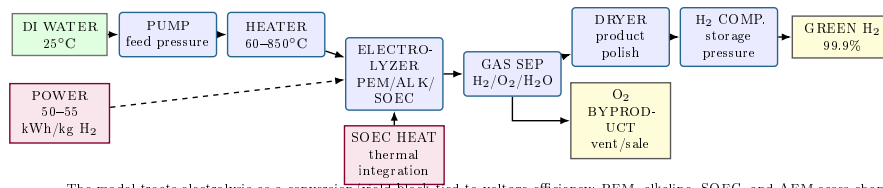
Parameter	Value	Unit
CNG dispenser cost	\$7.90	/MMBtu
CNG vs diesel saving	75–80	%
Well-to-wheel CO ₂ (no leak)	7.5	kg/GJ
Well-to-wheel CO ₂ (2% leak)	10.5	kg/GJ
Compression efficiency	70	%
		isothermal

Chapter 7

Water Electrolysis for Green Hydrogen

7.1 Aspen Plus Model

The electrolytic hydrogen model treats the stack as a conversion block driven by electricity demand and voltage efficiency, with upstream water conditioning and downstream gas separation, drying, and compression. PEM, alkaline, SOEC, and AEM cases use the same flowsheet structure while changing temperature, efficiency, current-density, and CAPEX inputs.



The model treats electrolysis as a conversion/yield block tied to voltage efficiency; PEM, alkaline, SOEC, and AEM cases change temperature, efficiency, and CAPEX inputs.

Figure 7.1: Aspen Plus electrolysis hydrogen model flowsheet

7.2 Cell Voltage Equation

$$V_{\text{cell}} = E_{\text{rev}} + \eta_{\text{act}} + \eta_{\text{ohm}} + \eta_{\text{conc}} \quad (7.1)$$

$$E_{\text{rev}} = 1.23 - 0.00084 \times (T - 298.15) \quad (7.2)$$

7.3 PEM vs. Alkaline Comparison

Table 7.1: PEM vs. Alkaline Electrolysis

Parameter	Alkaline	PEM
TRL	9	8–9
Temperature	60–90°C	50–80°C
Current density	0.2–0.8 A/cm ²	1.0–3.0 A/cm ²
System efficiency	60–70%	60–68%
CAPEX (\$/kW)	\$800–1,500	\$1,200–2,500

7.4 Solid Oxide Electrolysis (SOEC)

7.4.1 SOEC Principles

SOEC operates at high temperatures (600–1000°C) with solid zirconia-based electrolytes, enabling:

- Electrical efficiency up to 90% (LHV) with thermal integration
- Co-electrolysis of H₂O and CO₂ for syngas production
- Potential for >100 MW scales by 2030

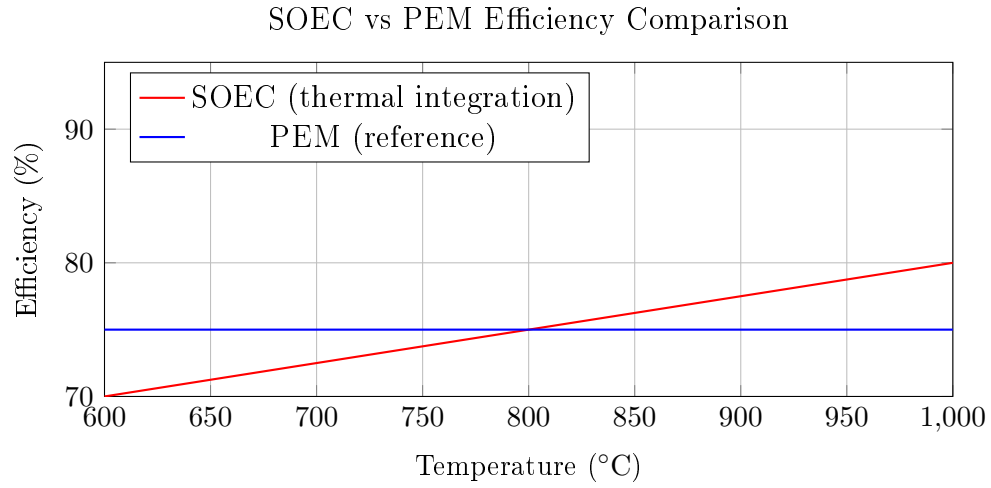


Figure 7.2: SOEC vs PEM Efficiency Comparison

Table 7.2: SOEC Performance Parameters

Parameter	Current	2030 Target
TRL	6–7	9
Temperature range	700–850°C	600–800°C
Current density	0.5–1.5 A/cm ²	1.5–3.0 A/cm ²
System efficiency	80–85%	85–90%
CAPEX (\$/kW)	\$2,500–4,500	\$800–1,500
Lifetime (hours)	10,000– 20,000	40,000– 60,000

7.5 Anion Exchange Membrane (AEM)

7.5.1 AEM Electrolysis

AEM combines benefits of alkaline and PEM:

- No precious metals (low-cost catalysts)
- Operating temperature 40–80°C
- Current density 0.5–2.0 A/cm²
- TRL 5–6, commercial expected 2028–2030

Table 7.3: AEM Electrolysis Status

Parameter	Current	2030 Target
TRL	5–6	8–9
Current density	0.5–1.0 A/cm ²	1.0–2.0 A/cm ²
System efficiency	60–65%	70–75%
CAPEX (\$/kW)	\$1,500–2,500	\$500–800

7.6 Key Results

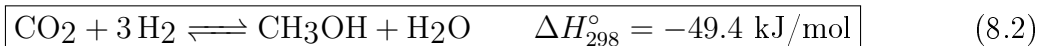
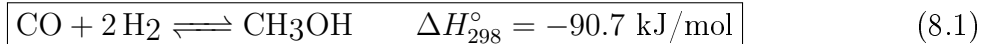
Table 7.4: Green Hydrogen Key Results

Parameter	Value	Unit
LCOH (China grid)	\$2.92	/kg
LCOH (EU grid)	\$7.50	/kg
LCOH (US grid)	\$5.20	/kg
LCOH (off-grid PV)	\$10.77	/kg
LCOH (hybrid PV+wind)	\$3.55	/kg
LCOH (SOEC, 2025)	\$4.80	/kg
LCOH (SOEC, 2030)	\$2.10	/kg
Water (wet cooling)	35	L/GJ
Water (dry cooling)	10	L/GJ
CAPEX (PEM, 100 MW)	\$1,400	/kW

Chapter 8

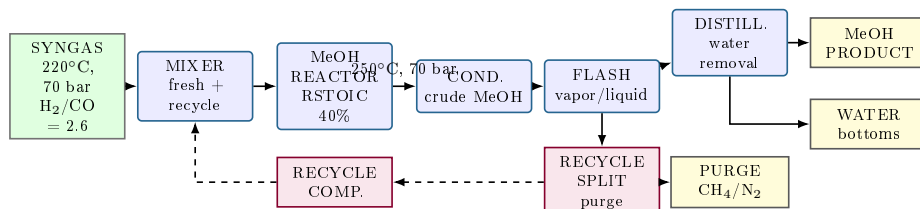
Methanol Production Pathways

8.1 Methanol Synthesis Reactions



8.2 Aspen Plus Model

The methanol Aspen Plus model uses a syngas mixer, RSTOIC synthesis reactor, condenser, flash separator, recycle loop, purge, and distillation column. This structure supports the conventional, biomass, blue methanol, and e-methanol cases by changing the syngas source while preserving the synthesis and purification backbone.



Model basis: syngas feed from natural gas, biomass, or captured CO₂ plus H₂; recycle improves single-pass conversion while purge controls inerts.

Figure 8.1: Aspen Plus methanol synthesis and purification model flowsheet

8.3 Production Costs

Table 8.1: Methanol Production Costs

Pathway	Cost (\$/tonne)	CO ₂ (kg/GJ)	TRL
Conventional (NG, \$3)	170	4.0	9
Conventional (coal)	150	7.5	9
Blue methanol	320	2.0	7–8
Biomass	350	2.7	6–7
BeBOP (target)	<250	0.7	6
e-Methanol (industrial)	380	1.0	6
e-Methanol (DAC target)	280	-1.0	4–5

8.4 Methanol for Shipping and Marine Fuel

8.4.1 Marine Sector Overview

International shipping accounts for ~3% of global CO₂ emissions. Methanol offers a viable decarbonization pathway due to:

- Liquid at ambient conditions (simplified storage)
- Existing bunkering infrastructure adaptation
- 95% lower SOx and 80% lower NOx vs. heavy fuel oil (HFO)

Table 8.2: Marine Fuel Comparison

Fuel	Energy Density (MJ/kg)	Storage Temp (°C)	CO ₂ (kg/GJ)	SO _x (g/GJ)
HFO	40.0	30	77.0	1000
LNG	50.0	-162	56.0	2
Methanol	20.0	25	35.0	0.5
Green Methanol	20.0	25	2.0	0.5
Ammonia	18.6	-33	0.0	0

8.4.2 Green Methanol for Shipping

Table 8.3: Shipping Methanol Deployment Scenarios

Scenario	2030	2050
Methanol ships (vessels)	200–500	5,000–10,000
Demand (million tonnes/year)	5–10	50–100
Cost premium vs. HFO	2–3x	1.5–2x

```

1 TITLE 'Methanol Synthesis from Syngas'
2
3 FLOWSHEET
4   BLOCK METHANOL_REACTOR IN = SYNGAS
5   BLOCK CONDENSER IN = REACTOR_OUT
6   BLOCK RECYCLE_SPLIT IN = CONDENSER_VAPOR
7   BLOCK DISTILLATION IN = CRUDE_METHANOL
8
9 STREAMS
10  SYNGAS TEMP=220 PRES=70 MOLE-FLOW=10000
11    COMP H2=0.65 CO=0.25 CO2=0.05 CH4=0.03 N2=0.02
12
13 BLOCK METHANOL_REACTOR RSTOIC
14   PARAM TEMP=250 PRES=70
15   REAC METHANOL_SYN CO + 2 H2 -> METHANOL CONV=0.40

```

Listing 8.1: Methanol Synthesis Aspen Plus Input

Part III

COMPARATIVE ANALYSIS

Part III Overview

Comparative Analysis

Part III converts the individual production-pathway results into decision-ready comparisons. The purpose is to show where each fuel performs well, where it carries penalties, and which trade-offs matter most for deployment.

The comparison is organized around three core questions:

- **Cost:** Which pathway delivers the lowest levelized cost under current and near-term assumptions?
- **Environment:** Which pathway offers the strongest life cycle carbon, water, and pollutant performance?
- **Safety and deployment:** Which pathway can be handled, stored, transported, permitted, and scaled with acceptable risk?

The final chapter in this part combines these dimensions through MCDA. This is important because the lowest-cost option is not always the best deployment choice once carbon intensity, safety, infrastructure readiness, and technology maturity are considered together.

Chapter 9

Techno-Economic Analysis

9.1 Levelized Cost Comparison

Table 9.1: LCOH Results

Production Method	LCOH (\$/kg)	CAPEX (\$/kW)	Efficiency (%)
Grey SMR	1.82	1,500	81
Blue SMR+CCS	3.22	2,200	75
Blue SMR+45Q	2.59	2,200	75
PEM (China)	2.92	1,400	68
PEM (EU)	7.50	1,400	68
PV off-grid	10.77	2,000	25% CF
Hybrid PV+Wind	3.55	1,800	55% CF
SOEC (2025)	4.80	3,500	82
SOEC (2030 target)	2.10	1,200	88
DOE 2030 Target	1.00	400	75

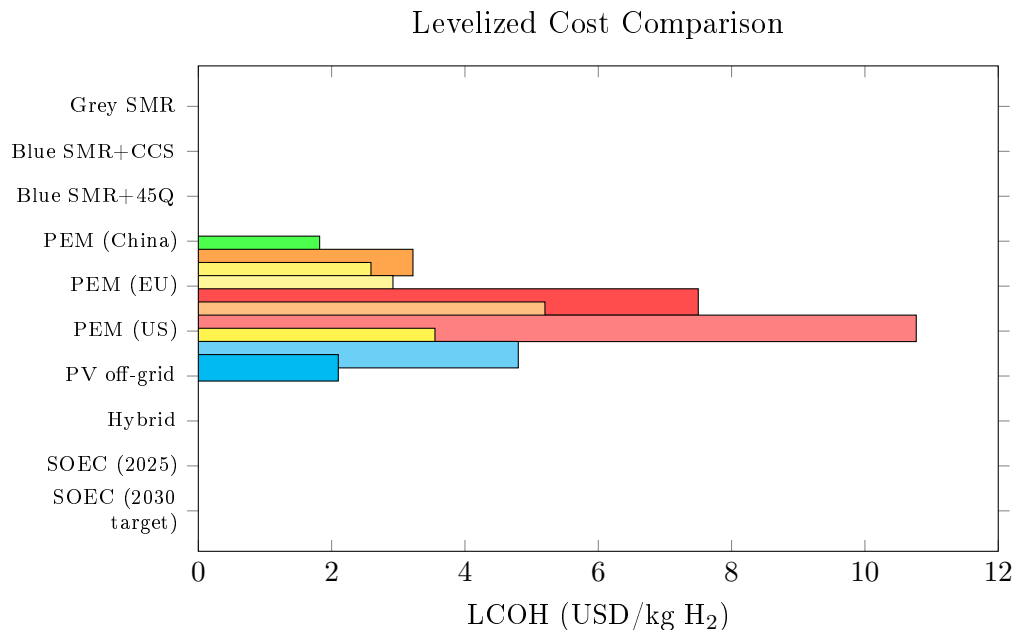


Figure 9.1: Levelized Cost Comparison

9.2 Graphical Cost–Carbon Portfolio

Cost alone does not identify the best pathway because the cheapest fuel may also lock in high emissions. Figure 9.2 places each pathway on a cost–carbon map so the strategic position of each option is visible at a glance. Grey SMR remains the lowest-cost baseline but sits in the high-carbon zone. Blue SMR reduces emissions but moves upward in cost

because of capture equipment, compression, storage, and energy penalties. Green hydrogen and e-methanol occupy the low-carbon region but require continued cost reduction and larger clean-electricity supply chains.

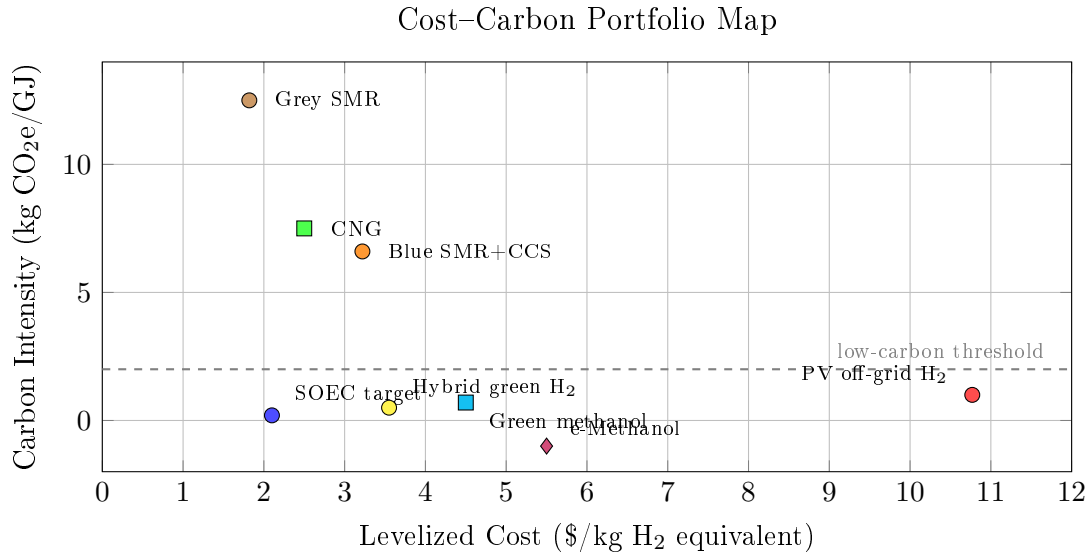


Figure 9.2: Cost–carbon portfolio map for selected engineered fuel pathways

The most attractive long-term quadrant is the lower-left region: low cost and low carbon intensity. Technologies move toward this region through three mechanisms: reduced electrolyzer CAPEX, higher capacity factor from hybrid renewable systems, and cheaper low-carbon CO₂ or biomass-derived carbon for liquid fuels. Transition fuels such as CNG remain useful where infrastructure already exists, but methane leakage control determines whether their climate benefit is preserved.

9.3 Cost Reduction Outlook

Figure 9.3 translates the cost comparison into a deployment timeline. The curves are scenario values rather than forecasts: they show what happens if technology learning, scale-up, higher utilization, and supply-chain maturity reduce costs at different rates. SOEC has the steepest potential decline because it begins from a higher early-stage cost base, while hybrid renewable hydrogen improves through better capacity factor and lower renewable electricity cost.

For investment planning, the important result is not a single 2050 number but the timing of cost convergence. If hybrid green hydrogen approaches \$2/kg before 2035, it becomes competitive for ammonia, refining, and some heavy-transport applications. If e-methanol remains above \$3/kg H₂-equivalent beyond 2035, its strongest early market is likely to remain premium shipping and aviation-fuel blending rather than general energy use.

9.4 CAPEX Analysis

9.5 Sensitivity Analysis

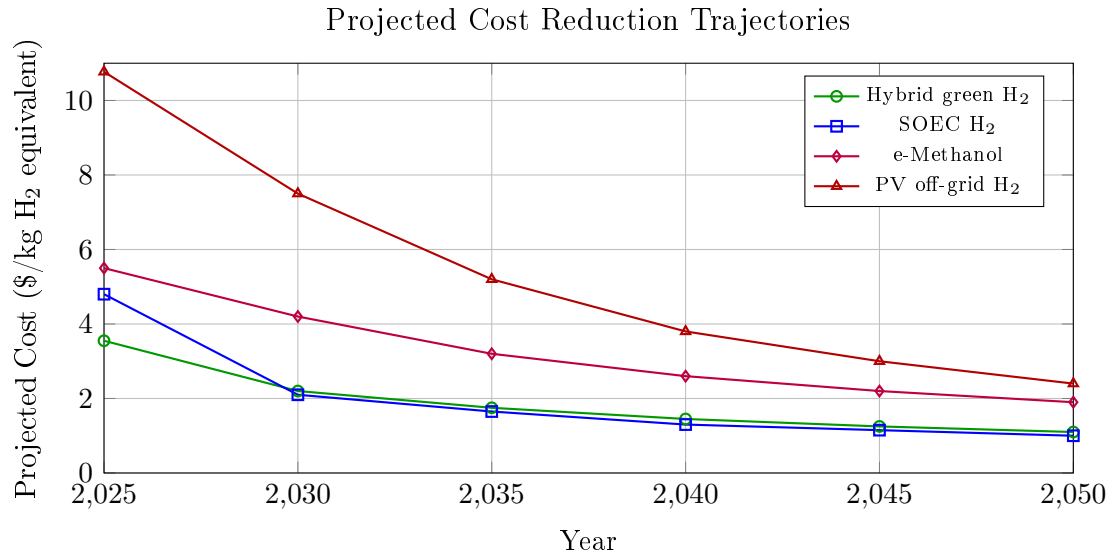


Figure 9.3: Illustrative cost reduction trajectories for major low-carbon pathways

Table 9.2: CAPEX Breakdown (\$/kW)

Technology	Equip.	Install	Eng.	Cont.	Total
SMR (100 tpd)	600	300	150	450	1,500
SMR+CCS	900	400	200	700	2,200
PEM (1 MW)	1,200	400	200	200	2,000
PEM (100 MW)	900	250	150	100	1,400
Alkaline (100 MW)	480	160	100	60	800
SOEC (1 MW)	2,200	600	300	400	3,500
SOEC (100 MW)	800	200	100	100	1,200
CNG Station	300	100	50	50	500

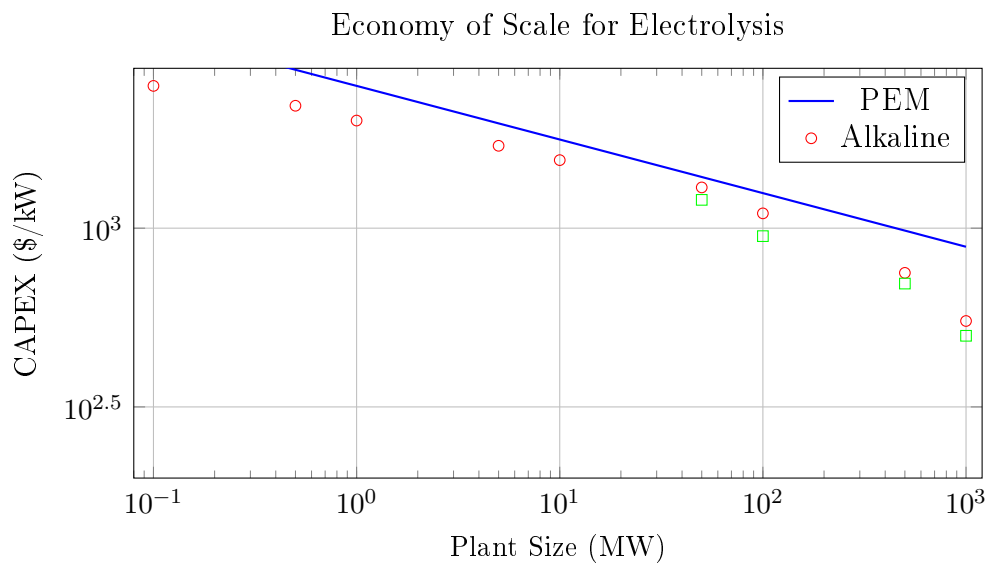


Figure 9.4: Economy of Scale for Electrolysis

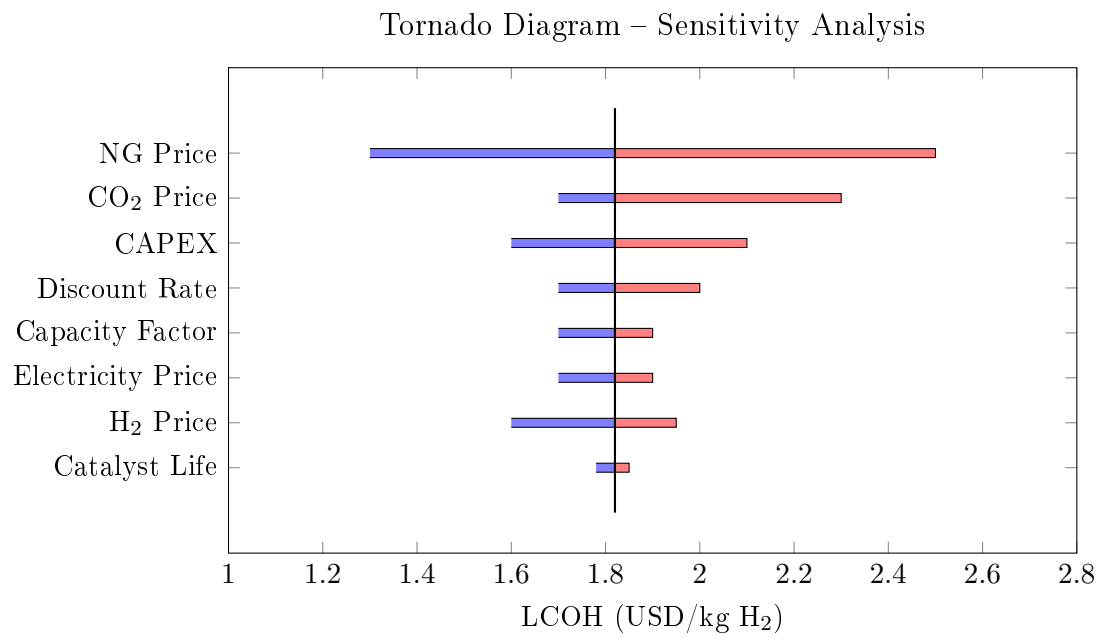


Figure 9.5: Tornado Diagram – SMR LCOH Sensitivity

Chapter 10

Environmental Impact Assessment

10.1 Full ISO 14040/14044 LCA Framework

10.1.1 Goal and Scope

The LCA follows ISO 14040:2006 and ISO 14044:2006 with:

- Functional unit: 1 GJ of fuel energy delivered to end user
- System boundary: Cradle-to-grave (extraction to end-use combustion)
- Allocation: Mass basis for co-products (e.g., export steam)
- Data sources: Ecoinvent v3.10, GREET 2024, literature values

10.1.2 Impact Categories

Table 10.1: LCIA Impact Categories and Methods

Impact Category	Method	Unit
Global warming (GWP100)	IPCC AR6	kg CO ₂ e
Acidification	ReCiPe 2016	kg SO ₂ e
Eutrophication	ReCiPe 2016	kg P equivalent
Water consumption	AWARE	m ³
Ozone depletion	ReCiPe 2016	kg CFC-11e
Photochemical smog	ReCiPe 2016	kg NMVOC

10.2 Carbon Footprint

Table 10.2: Life Cycle Carbon Footprint (kg CO₂e/GJ)

Fuel Pathway	Total CO ₂ e
Gasoline	11.0
Grey H ₂ (SMR)	12.5
Blue H ₂ (SMR+CCS)	6.6
Green H ₂ (PV)	1.0
CNG (no leak)	7.5
CNG (2% leak)	10.5
Methanol (NG)	4.0
Green Methanol	0.7
e-Methanol	-1.0

Figure 10.1 turns the carbon-footprint table into a visual screening chart. The negative value for e-methanol represents a pathway where captured carbon is credited at the system boundary; it should be interpreted carefully because the result depends on the permanence, source, and allocation method for captured CO₂.

The chart highlights two important deployment implications. First, CCS substantially improves SMR performance but does not make it equivalent to renewable electrolysis.

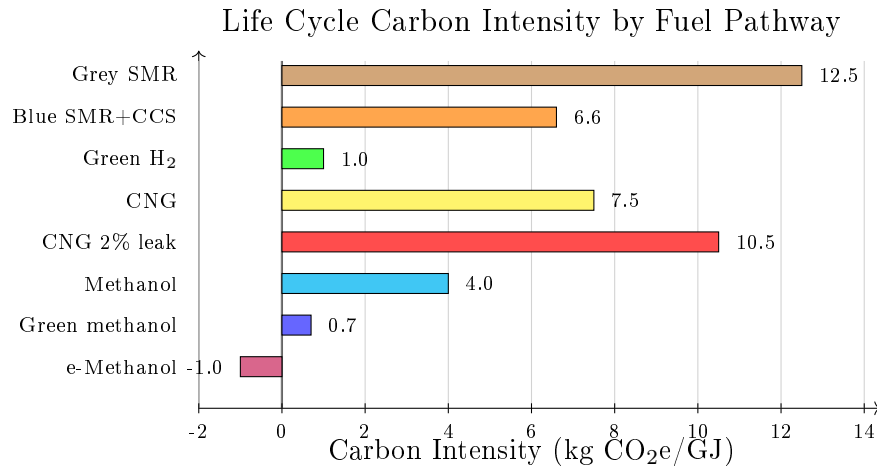


Figure 10.1: Graphical comparison of life cycle carbon intensity

Second, CNG remains sensitive to upstream methane leakage; small leakage rates can erase a large share of its climate advantage over liquid petroleum fuels.

10.3 Water Consumption

Table 10.3: Water Consumption (L/GJ)

Technology	Water
SMR H ₂	30
SMR+CCS	34
PEM (wet cooling)	35
PEM (dry cooling)	10
SOEC	8
CNG	0.5
Methanol (NG)	17

10.4 Air Pollutant Emissions

Table 10.4: Air Pollutant Emissions (g/GJ)

Technology	NO _x	PM _{2.5}	CO
SMR H ₂	15	2.0	5
PEM (coal grid)	30	2.5	0
CNG	10	0.5	15
Methanol	8	0.3	20
Gasoline	20	2.0	50

Chapter 11

Safety and Risk Analysis

11.1 Flammability Limits

Table 11.1: Flammability Properties

Fuel	LFL (vol%)	UFL (vol%)	MIE (mJ)
Hydrogen	4.0	75.0	0.018
Methane	5.0	15.0	0.29
Methanol	6.7	36.0	0.14
Gasoline	1.4	7.6	0.24

11.1.1 Screening Matrix

The flammability table gives physical properties, but operators need an immediate risk-screening view. Figure 11.1 maps qualitative likelihood against consequence for the major fuels. Hydrogen is placed in the high-consequence/high-likelihood corner because of its wide flammability range, low minimum ignition energy, high diffusivity, and high-pressure storage systems. Methanol sits lower on the matrix because it is a liquid at ambient conditions, although toxicity and spill control remain important.

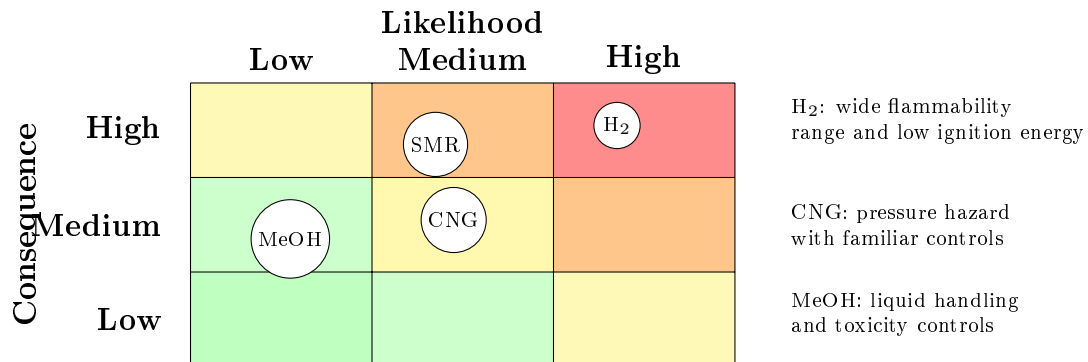


Figure 11.1: Safety screening matrix for major engineered fuel pathways

The matrix is not a substitute for full QRA, but it gives a clear first-pass view of where engineering controls should be strongest: hydrogen requires leak detection, ventilation, classified electrical equipment, pressure relief, and material-compatibility management; CNG requires pressure-vessel integrity and compressor safeguards; methanol requires spill containment, vapor control, and occupational exposure protection.

11.2 Storage System Mass

11.3 Quantitative Risk Assessment (QRA)

11.3.1 Risk Metrics

Table 11.2: Storage System Mass (5 kg H₂ equivalent)

Technology	Total Mass (kg)
Gasoline	20
Methanol	60
H ₂ (liquid)	100
CNG (250 bar)	120
H ₂ (700 bar)	150
H ₂ (350 bar)	200

Table 11.3: QRA Metrics by Pathway

Pathway	Fatality Rate (per GJ)	Injury Rate (per GJ)	Property Loss (\$/GJ)
SMR H ₂	1.2×10^{-7}	3.5×10^{-6}	0.15
CNG	8.5×10^{-8}	2.1×10^{-6}	0.08
Methanol	4.2×10^{-8}	1.3×10^{-6}	0.05
Green H ₂	9.8×10^{-8}	2.8×10^{-6}	0.12

11.4 ATEX and HAZOP Analysis

11.4.1 ATEX Zones for Hydrogen

Table 11.4: ATEX Classification for Hydrogen Systems

Zone	Duration	Location	Equipment
Zone 0	Continuous	Inside enclosure	Ex ia
Zone 1	Intermittent	Ventilation points	Ex d
Zone 2	Occasional	Piping and storage	Ex n

11.4.2 HAZOP Study Results

11.5 Dispersion Modeling

Table 11.5: HAZOP Study for SMR Plant

Node	Deviation	Cause	Risk Level
Reformer inlet	High temperature	Burner failure	High
HTS outlet	Low pressure	Valve failure	Medium
PSA	Low H ₂ purity	Adsorbent degradation	Medium

Table 11.6: Dispersion Characteristics (CFD Modeling)

Fuel	Plume Rise (m)	Flammable Cloud (m ³)	Time to LFL (s)
Hydrogen	15–25	50–80	0.5–1.0
Methane	10–15	100–150	1.0–2.0
Methanol	5–10	200–300	2.0–3.0

Chapter 12

Multi-Criteria Decision Analysis

12.1 MCDA Results

Table 12.1: MCDA Scores and Rankings

Technology	Cost	Carbon	Safety	Infra.	TRL	Scale	Score
Green Methanol	0.60	0.95	0.80	0.70	0.70	1.00	0.80
Methanol (NG)	0.60	0.70	0.80	0.90	1.00	0.60	0.73
CNG	0.80	0.50	0.60	0.90	1.00	0.60	0.70
Green H ₂	0.40	0.95	0.20	0.30	0.90	0.80	0.61
Grey H ₂	0.67	0.10	0.50	0.90	1.00	0.40	0.52

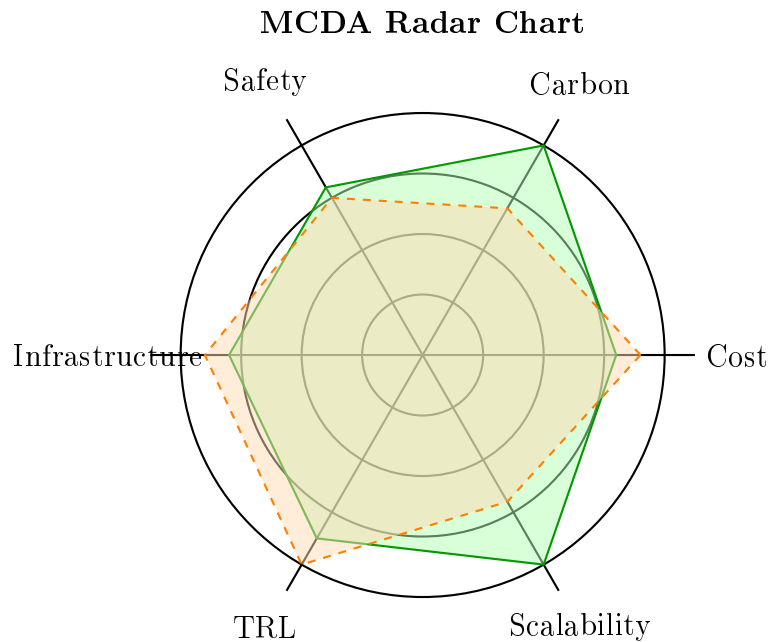


Figure 12.1: Multi-Criteria Decision Analysis Radar Chart

12.2 Weighting Methodology and Stakeholder Analysis

12.2.1 Weighting Scenario Comparison

Table 12.2: MCDA Scores by Stakeholder Scenario

Technology	Govt.	Industry	NGO	Cons.
Green Methanol	0.82	0.78	0.85	0.75
CNG	0.68	0.72	0.60	0.72
Green H ₂	0.65	0.58	0.78	0.52
Grey H ₂	0.48	0.55	0.30	0.60

12.2.2 Sensitivity Analysis

Table 12.3: MCDA Sensitivity to Weighting Changes

Technology	Equal Weights	Cost- Weighted	Carbon- Weighted
Green Methanol	0.78	0.72	0.85
CNG	0.72	0.78	0.62
Green H ₂	0.62	0.52	0.78
Grey H ₂	0.52	0.58	0.38

Part IV

ADVANCED TOPICS

Part IV Overview

Advanced Topics

Part IV extends the main comparison into emerging questions that affect real-world adoption. These chapters address practical edge cases, technology uncertainty, regional deployment barriers, and future pathways that may change the ranking of fuels over time.

The advanced topics are included for three reasons:

- They test whether the conclusions remain valid when water quality, pipeline blending, hydrogen leakage, synthetic fuels, or sector-specific demand are considered.
- They identify innovation areas such as SOEC, AEM electrolysis, e-SMR, advanced carbon capture, direct-air-capture methanol, and biomass-derived methanol.
- They connect global technology choices to regional deployment, especially where renewable resources, infrastructure gaps, finance, and policy capacity differ sharply.

This part is deliberately more exploratory than Parts I–III. It helps distinguish mature recommendations from questions that require monitoring, pilot projects, or further modelling before large-scale investment decisions.

Chapter 13

Curious Research Questions & Solutions

13.1 Q1: Hydrogen from Wastewater?

Answer: Yes, with 3–5× higher pretreatment energy. Net cost increase: \$0.15/kg H₂.

Table 13.1: Water Source Comparison

Parameter	Fresh	Municipal WW	Industrial WW	Seawater
Pretreatment energy (kWh/m ³)	0.5	2.5	4.0	3.5
Additional H ₂ cost (\$/kg)	0	0.08	0.15	0.12

13.2 Q2: Hydrogen Blending in Pipelines?

Answer: Up to 10% H₂ is safe. Above 20%, significant retrofit required.

Table 13.2: Pipeline H₂ Compatibility

Steel Grade	Max Safe H ₂ Blend (%)	Failure Mode
X42	30%	Ductile fracture
X52	20%	Brittle fracture
X65	10%	Subcritical cracking
X70	5%	Rapid crack growth
Cast iron	0%	Graphitization

13.3 Q3: e-Methanol from DAC?

Answer: Not yet. Breakeven requires carbon price > \$136/tonne.

$$\begin{aligned}\text{Breakeven carbon price} &= \frac{\$190}{1.4} \\ &= \$136/\text{tonne CO}_2\end{aligned}\tag{13.1}$$

13.4 Q4: Land for Green Hydrogen?

Answer: 660,000 km² (0.44% of land area) with wind+solar hybrid.

Table 13.3: Land Requirements

Scenario	Land Area (km ²)
100% H ₂ from solar	3,000,000
100% H ₂ from wind	1,000,000
50% solar + 50% wind	660,000

13.5 Q5: Climate Cost of H₂ Leakage?

Answer: GWP(100-year) = 10, GWP(20-year) = 33.

$$\text{GWP}_{\text{H}_2} = 28 \times \frac{12}{2} \times 0.5 \times 0.125 = 10.5 \quad (13.2)$$

13.6 Q6: Abandoned Oil Wells for H₂ Storage?

Answer: Depleted gas reservoirs: yes. Oil wells: no.

Table 13.4: Geological Storage Suitability

Reservoir Type	Suitability	Capacity (t H ₂ /m ³)
Depleted gas (sandstone)	High	15–20
Salt cavern	Very High	20–30
Depleted oil	Low	5–10

13.7 Q7: Tire Particulate Emissions?

Answer: BEV (21.5 mg/km) < Gasoline (24) < Diesel (34).

Table 13.5: PM2.5 Emissions by Vehicle (mg/km)

Vehicle	Tire	Brake	Exhaust
Gasoline	10	4	5
Diesel	10	4	15
BEV (1,800 kg)	14	0.5	0
Heavy BEV (2,400 kg)	21	0.5	0

13.8 Q8: Seawater Electrolysis?

Answer: Not yet. Desalinate first (adds \$0.02/kg H₂).

13.9 Q9: Minimum Energy for Synthetic Fuel?

Answer: Theoretical: 4.4 kWh/kg methanol. Current: 9.4 kWh/kg.

13.10 Q10: Hybrid SMR + Electrolysis?

Answer: Yes – optimal when electricity < 2.67¢/kWh.

13.11 Q11: Ammonia as Hydrogen Carrier?

Answer: Yes – 17.6 wt% H₂ but requires cracking (15–20% energy penalty).

Table 13.6: Energy Requirements (kWh/kg)

Process	Theoretical	Practical	Current
e-methanol (PEM)	6.1	7.5	9.4
e-methanol (SOEC)	4.4	5.5	7.0

Table 13.7: Hybrid Optimization

Electricity Price (\$/kWh)	Optimal H ₂ from Elec (%)	LCOH (\$/kg)
0.01	100%	1.20
0.02	85%	1.45
0.03	40%	1.65
0.04	0%	1.82

13.12 Q12: Hydrogen for Steelmaking?

Answer: DRI with H₂ reduces CO₂ by 90% at 2–3x current cost.

13.13 Q13: Hydrogen for Cement?

Answer: 50% H₂ in kiln reduces CO₂ by 35% (process emissions remain).

$$\begin{aligned}\text{CO}_2 \text{ reduction} &= 0.50 \times 0.65 \\ &= 0.325 \text{ (32.5\%)}\end{aligned}\tag{13.3}$$

13.14 Q14: Hydrogen for Aviation?

Answer: Liquid H₂ possible but requires 4x tank volume vs. kerosene.

13.15 Q15: Hydrogen for Heating?

Answer: 20% blend safe, 100% requires new appliances (efficiency 95–98%).

13.16 Q16: Hydrogen for Power Generation?

Answer: 30–40% efficiency in gas turbines vs. 45–50% for natural gas.

13.17 Q17: Hydrogen for Agriculture (Fertilizer)?

Answer: 100% green H₂ for Haber-Bosch adds \$200–300/tonne NH₃.

Table 13.8: Hydrogen Carriers Comparison

Carrier	H ₂ content (wt%)	Energy penalty (%)	TRL
Ammonia	17.6	15–20	7–8
LOHC (toluene)	6.2	30–35	6–7
Methanol	12.6	10–15	8–9
Liquid H ₂	100	25–30	7–8

Table 13.9: Hydrogen in Steelmaking

Parameter	Blast Furnace	H₂-DRI
CO ₂ (kg/kg steel)	1.8	0.15
Energy consumption (GJ/t)	14	8
H ₂ demand (kg/t steel)	0	50
Cost premium (%)	0%	100–200%

Table 13.10: Aviation Fuel Comparison

Parameter	Kerosene	Liquid H₂
Energy density (MJ/kg)	43	120
Volumetric density (MJ/L)	35	8.5
Tank volume ratio	1.0	4.1
CO ₂ emissions (kg/GJ)	71	0

Table 13.11: Power Generation Comparison

Parameter	NG Combined Cycle	H₂ Combined Cycle
Efficiency (%)	45–50	30–40
NOx emissions (ppm)	10–25	5–15 (with dilution)
CAPEX (\$/kW)	600–800	1,200–1,800

$$\begin{aligned}\text{NH}_3 \text{ cost increase} &= \frac{3 \times 2.016}{17.03} \times \$2.92 \\ &= \$1.04/\text{kg NH}_3\end{aligned}\tag{13.4}$$

13.18 Q18: Hydrogen for Refining?

Answer: 50% of refinery H₂ can be green by 2035 (cost premium \$0.50/kg).

13.19 Q19: Hydrogen for Food Processing?

Answer: Edible grade H₂ requires \$0.20/kg additional purification.

13.20 Q20: Hydrogen for Medical Applications?

Answer: Medical grade H₂ (99.9995%) costs \$8–12/kg (\$10–15/kg premium).

Table 13.12: Hydrogen Grades and Costs

Grade	Purity (%)	Cost (\$/kg)	Application
Industrial	99.9	1.5–3.0	Refining, chemical
Fuel cell	99.99	3.0–6.0	Transportation
Medical	99.9995	8.0–12.0	Medical imaging
Semiconductor	99.99999	15.0–25.0	Electronics

Chapter 14

Emerging Technologies & Future Pathways

14.1 SMR Innovations

Table 14.1: SMR Emerging Technologies

Technology	TRL	Commercial Scale
Electrified SMR (e-SMR)	5–6	2028–2030
Sorption-Enhanced SMR	5	2030+
Biogas SMR	7–8	Deployed

14.2 Electrolysis Innovations

Table 14.2: Electrolysis Emerging Technologies

Technology	TRL	Availability
Modular PEM (truck-deployable)	7–8	2026
Anion Exchange Membrane (AEM)	5–6	2028–2030
Solid Oxide (SOEC)	6–7	2030

14.3 Methanol Innovations

Table 14.3: Methanol Emerging Technologies

Technology	TRL	Commercial Scale
BeBOP (SOEC-biomass)	6	2028–2030
Leuna100 homogeneous cat.	6	2027
Methanol-to-jet (ATJ)	7–8	2026

14.4 Carbon Capture Innovations

Table 14.4: Carbon Capture Emerging Technologies

Technology	TRL	Availability
Direct Air Capture (DAC)	6–7	2027–2030
Bioenergy with CCS (BECCS)	6–7	2028+
Calcium looping	6	2028–2030

Chapter 15

Regional Deployment Strategies

15.1 Regional Suitability

Table 15.1: Regional Suitability (1–5 scale)

Region	CNG	Blue H ₂	Green H ₂	Methanol
North America	5	5	4	4
Europe	4	4	5	5
China	4	4	5	5
India	5	3	4	3
Middle East	3	4	5	4
Australia	3	3	5	3
Sub-Saharan Africa	2	2	5	3
Latin America	3	3	4	3

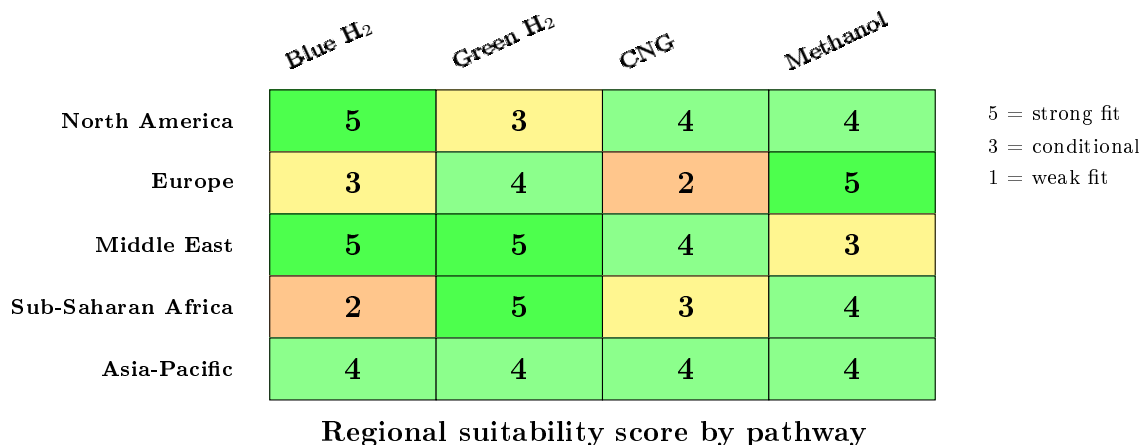


Figure 15.1: Regional deployment heatmap for major engineered fuel pathways

The heatmap gives a quick deployment screen that complements the table. Blue hydrogen scores strongly in regions with natural gas resources, CO₂ storage capacity, and existing industrial hydrogen demand. Green hydrogen performs best where high-quality renewable resources can be paired with export infrastructure or domestic industrial demand. Methanol receives high scores in regions with ports, chemical infrastructure, and maritime fuel demand, while CNG is strongest where gas distribution and vehicle-fuelling infrastructure already exist.

15.2 Sub-Saharan Africa and Emerging Markets

15.2.1 Africa's Hydrogen Potential

Sub-Saharan Africa has exceptional renewable energy resources:

- Solar PV capacity factor: 25–35% (Sahara region)

- Wind capacity factor: 35–45% (coastal regions)
- Hydroelectric potential: 300+ GW (Congo River basin)

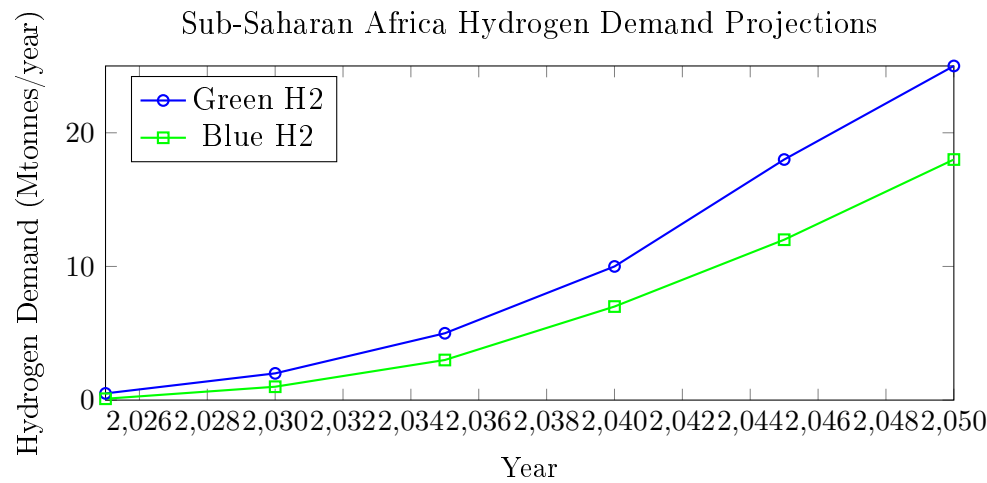


Figure 15.2: Sub-Saharan Africa Hydrogen Demand Projections

15.2.2 Deployment Barriers and Solutions

Table 15.2: Africa Hydrogen Deployment Barriers

Barrier	Impact	Mitigation Strategy
Infrastructure gap	High CAPEX	International partnerships
Water scarcity	10–15% cost premium	Desalination with renewable
Grid instability	20–30% efficiency loss	Off-grid hybrid systems
Policy uncertainty	15–20% risk premium	Regulatory frameworks

15.2.3 International Partnerships

Table 15.3: Proposed Africa-Europe Hydrogen Partnerships

Partner	Target (2030)	Investment (\$B)
Germany-Namibia	2 Mt/year	10
EU-Morocco	1 Mt/year	8
Japan-Egypt	1 Mt/year	6
UK-South Africa	0.5 Mt/year	4

15.3 Technology Roadmap to 2050

Table 15.4: Deployment Roadmap

Technology	2025–2030	2030–2040	2040–2050
CNG	Deploy	Transition to R-CNG	Phase out
Blue H ₂	Scale with CCS	Peak deployment	Transition to green
Green H ₂	Demonstration	Commercial scale	Dominant
Methanol	Fossil + green blends	Green methanol scales	e-methanol

Part V

REFERENCE MATERIALS

Part V Overview

Reference Materials

Part V gathers the equations, example calculations, model inputs, plotting code, data tables, glossary, and nomenclature needed to reproduce or extend the analysis. It is designed as a working reference for readers who want to check assumptions, reuse formulas, or adapt the report to another project.

- The equation compendium consolidates thermodynamic, kinetic, electrochemical, economic, and fluid-mechanics formulas.
- Worked examples show how the equations are applied to reactor sizing, electrolyzer design, LCOH estimation, methanol synthesis, and CNG compression.
- Aspen Plus input files and Python scripts provide starting points for simulation and figure generation.
- The appendices, glossary, and nomenclature tables standardize terminology and symbols used across the full document.

This reference section makes the report more than a narrative comparison: it becomes a reusable technical package for future fuel-pathway studies.

Chapter 16

Complete Equation Compendium

16.1 Thermodynamics

$$\boxed{\Delta U = Q - W} \quad (\text{First Law}) \quad (16.1)$$

$$\boxed{H = U + PV} \quad (\text{Enthalpy}) \quad (16.2)$$

$$\boxed{dS = \frac{\delta Q_{\text{rev}}}{T}} \quad (\text{Entropy}) \quad (16.3)$$

$$\boxed{G = H - TS} \quad (\text{Gibbs Free Energy}) \quad (16.4)$$

$$\boxed{Ex = (h - h_0) - T_0(s - s_0)} \quad (\text{Exergy}) \quad (16.5)$$

$$\boxed{PV = nRT} \quad (\text{Ideal Gas}) \quad (16.6)$$

$$\boxed{\eta_{\text{Carnot}} = 1 - \frac{T_{\text{cold}}}{T_{\text{hot}}}} \quad (\text{Carnot}) \quad (16.7)$$

16.2 Equations of State

$$\boxed{P = \frac{RT}{V - b} - \frac{a\alpha}{V(V + b) + b(V - b)}} \quad (\text{Peng-Robinson}) \quad (16.8)$$

$$a = 0.45724 \frac{R^2 T_c^2}{P_c} \quad (16.9)$$

$$b = 0.07780 \frac{RT_c}{P_c} \quad (16.10)$$

$$\alpha = \left[1 + \kappa \left(1 - \sqrt{T_r} \right) \right]^2 \quad (16.11)$$

$$\kappa = 0.37464 + 1.54226\omega - 0.26992\omega^2 \quad (16.12)$$

16.3 Kinetics

$$k = A \exp\left(-\frac{E_a}{RT}\right) \quad (\text{Arrhenius}) \quad (16.13)$$

$$r = k C_A^\alpha C_B^\beta \quad (\text{Power Law}) \quad (16.14)$$

$$r = \frac{k K_A P_A K_B P_B}{(1 + K_A P_A + K_B P_B)^2} \quad (\text{Langmuir-Hinshelwood}) \quad (16.15)$$

$$r_1 = \frac{k_1}{P_{\text{H}_2}^{2.5}} \cdot \frac{P_{\text{CH}_4} P_{\text{H}_2\text{O}} - \frac{P_{\text{H}_2}^3 P_{\text{CO}}}{K_1}}{\text{DEN}^2} \quad (\text{Xu-Froment}) \quad (16.16)$$

16.4 Electrochemistry

$$E = E^\circ - \frac{RT}{nF} \ln Q \quad (\text{Nernst}) \quad (16.17)$$

$$j = j_0 \left[\exp\left(\frac{\alpha n F \eta}{RT}\right) - \exp\left(-\frac{(1 - \alpha) n F \eta}{RT}\right) \right] \quad (\text{Butler-Volmer}) \quad (16.18)$$

$$\eta = a + b \log(j) \quad (\text{Tafel}) \quad (16.19)$$

$$\dot{m}_{\text{H}_2} = \frac{I \times M_{\text{H}_2}}{n \times F} \times \eta_{\text{Faraday}} \quad (\text{Faraday's Law}) \quad (16.20)$$

16.5 Economics

$$\text{LCOH} = \frac{\text{CAPEX} \times \text{CRF} + \text{OPEX}_{\text{fixed}} + \text{Feedstock}_{\text{cost}} \times \text{Annual}_{\text{consumption}}}{\text{Annual}_{\text{H}_2}} \quad (16.21)$$

$$\text{CRF} = \frac{r(1+r)^n}{(1+r)^n - 1} \quad (16.22)$$

$$\text{NPV} = \sum_{t=0}^n \frac{CF_t}{(1+r)^t} \quad (16.23)$$

$$C = C_0 \left(\frac{Q}{Q_0} \right)^{-b} \quad (\text{Learning Curve}) \quad (16.24)$$

16.6 Fluid Mechanics

$$\boxed{\frac{P_1}{\rho g} + \frac{v_1^2}{2g} + z_1 = \frac{P_2}{\rho g} + \frac{v_2^2}{2g} + z_2 + h_L} \quad (\text{Bernoulli}) \quad (16.25)$$

$$\boxed{\Delta P = f \frac{L}{D} \frac{\rho v^2}{2}} \quad (\text{Darcy-Weisbach}) \quad (16.26)$$

$$\boxed{\frac{\Delta P}{L} = 150 \frac{(1 - \varepsilon)^2}{\varepsilon^3} \frac{\mu u}{d_p^2} + 1.75 \frac{1 - \varepsilon}{\varepsilon^3} \frac{\rho u^2}{d_p}} \quad (\text{Ergun}) \quad (16.27)$$

$$\boxed{W = \frac{\gamma}{\gamma - 1} R T_1 \left[\left(\frac{P_2}{P_1} \right)^{\frac{\gamma - 1}{\gamma}} - 1 \right]} \quad (\text{Compressor Work}) \quad (16.28)$$

Chapter 17

Worked Calculation Examples

17.1 Example 1: SMR Reactor Design

Problem: Design a steam methane reformer to produce 100 tpd of hydrogen.

Step 1: Material Balance

$$\begin{aligned}\text{CH}_4 \text{ required} &= \frac{2,083}{4} \times 1.10 \\ &= 573 \text{ kmol/hr}\end{aligned}\tag{17.1}$$

$$\begin{aligned}\text{Steam} &= 573 \times 3.0 \\ &= 1,719 \text{ kmol/hr}\end{aligned}\tag{17.2}$$

Step 2: Catalyst Volume

$$\begin{aligned}V_{\text{gas}} &= 2,292 \times 22.4 \times \frac{1123}{273} \times \frac{1.013}{20} \\ &= 10,700 \text{ m}^3/\text{hr}\end{aligned}\tag{17.3}$$

$$\begin{aligned}V_{\text{catalyst}} &= \frac{10,700}{5,000} \\ &= 2.14 \text{ m}^3\end{aligned}\tag{17.4}$$

Step 3: Number of Tubes

$$\begin{aligned}\text{Volume per tube} &= \pi \times (0.05)^2 \times 10 \\ &= 0.0785 \text{ m}^3\end{aligned}\tag{17.5}$$

$$\begin{aligned}\text{Number of tubes} &= \frac{2.14}{0.0785} \\ &\approx 27 \text{ tubes (use 30)}\end{aligned}\tag{17.6}$$

17.2 Example 2: PEM Electrolyzer Sizing

Problem: Design a 1 MW PEM electrolyzer stack.

Step 1: Current and Cells

$$\begin{aligned}I &= \frac{1,000,000}{750} \\ &= 1,333 \text{ A}\end{aligned}\tag{17.7}$$

54

$$\begin{aligned} N_{\text{cells}} &= \frac{750}{1.9} \\ &= 395 \text{ cells} \end{aligned} \quad (17.8)$$

Step 2: H₂ Production

$$\begin{aligned} \dot{m}_{\text{H}_2} &= \frac{1,333 \times 395 \times 2.016}{2 \times 96,485} \times 0.95 \\ &= 18.8 \text{ kg/hr} \end{aligned} \quad (17.9)$$

17.3 Example 3: LCOH Calculation

Problem: Calculate LCOH for grey SMR.

$$\begin{aligned} \text{CRF} &= \frac{0.07(1.07)^{25}}{(1.07)^{25} - 1} \\ &= 0.0858 \end{aligned} \quad (17.10)$$

$$\begin{aligned} \text{LCOH} &= \frac{1,287,000 + 52,250,000 + 750,000}{35,000} \\ &= \$1.55/\text{kg} \end{aligned} \quad (17.11)$$

17.4 Example 4: Methanol Synthesis

Problem: Design a methanol reactor for 100,000 tonnes/year.

$$\begin{aligned} \text{Methanol flow} &= \frac{100,000,000}{32.04 \times 8,000} \\ &= 390 \text{ kmol/hr} \end{aligned} \quad (17.12)$$

$$\begin{aligned} F &= \frac{390}{0.115} \\ &= 3,391 \text{ kmol/hr fresh syngas} \end{aligned} \quad (17.13)$$

$$\begin{aligned} \text{Recycle ratio} &= \frac{2,093}{3,391} \\ &= 0.62 \end{aligned} \quad (17.14)$$

17.5 Example 5: CNG Compression

Problem: Design a three-stage CNG compressor for 1,000 kg/hr.

$$\begin{aligned}\text{Stage ratio} &= \left(\frac{200}{5}\right)^{1/3} \\ &= 3.42\end{aligned}\tag{17.15}$$

$$\begin{aligned}W_{\text{stage}} &= \frac{4.226 \times 2,478 \times 0.35}{0.85} \\ &= 4,306 \text{ kJ/kmol}\end{aligned}\tag{17.16}$$

$$\begin{aligned}P_{\text{total}} &= 63.6 + 66.8 + 66.8 \\ &= 197.2 \text{ kW}\end{aligned}\tag{17.17}$$

Chapter 18

Aspen Plus Complete Input Files

```
1 TITLE 'SMR Hydrogen Production - Base Case'
2
3 FLOWSHEET
4   BLOCK REFORMER IN = NG_FEED STEAM
5   BLOCK HTS IN = REFORMER_OUT
6   BLOCK LTS IN = HTS_OUT
7   BLOCK PSA IN = LTS_OUT
8
9 COMPONENTS
10  H2 HYDROGEN / C1 METHANE / CO CARBON-MONOXIDE /
11  CO2 CARBON-DIOXIDE / H2O WATER / N2 NITROGEN
12
13 PROPERTIES PENG-ROB
14   FREE-WATER STEAMNBS
15
16 STREAMS
17   NG_FEED TEMP=25 PRES=20 MOLE-FLOW=1000
18     COMP C1=0.92 C2=0.045 N2=0.01 CO2=0.005
19   STEAM TEMP=250 PRES=25 MOLE-FLOW=3000 COMP H2O=1
20
21 BLOCK REFORMER PFR
22   PARAM LENGTH=10 DIAMETER=1.5 NTUBES=100
23   REACTIONS SMR-KINETICS
24   SPEC TEMP-SPEC PROFILE "TEMP_PROFILE.TXT"
25
26 BLOCK HTS RSTOIC
27   PARAM TEMP=400 PRES=19.5
28   REAC WGS CO + H2O -> CO2 + H2 CONV=0.85
29
30 BLOCK LTS RSTOIC
31   PARAM TEMP=210 PRES=19
32   REAC WGS CO + H2O -> CO2 + H2 CONV=0.95
33
34 BLOCK PSA SEP
35   SPEC FRAC H2=0.9999 IN H2_PRODUCT
36   PARAM RECOVERY=0.85
```

Listing 18.1: SMR_BaseCase.apw (complete)

```
1 Length_Fraction, Temperature_C
2 0.000, 600
3 0.100, 645
4 0.200, 690
5 0.300, 728
6 0.400, 760
7 0.500, 788
8 0.600, 812
9 0.700, 833
10 0.800, 845
11 0.900, 850
12 1.000, 850
```

Listing 18.2: TEMP_PROFILE.txt

```
1 TITLE 'CNG Conditioning and Compression - Base Case'
2
3 FLOWSHEET
4   BLOCK INLET_SEP IN = RAW_GAS
5   BLOCK DEHYDRATOR IN = DRYER_FEED
6   BLOCK COMP1 IN = DRY_GAS
```

```

7  BLOCK IC1 IN = COMP1_OUT
8  BLOCK COMP2 IN = IC1_OUT
9  BLOCK IC2 IN = COMP2_OUT
10 BLOCK COMP3 IN = IC2_OUT
11 BLOCK AFTERCOOLER IN = COMP3_OUT
12 BLOCK STORAGE IN = CNG_PRODUCT
13 BLOCK DISPENSER IN = STORAGE_OUT
14
15 COMPONENTS
16   C1 METHANE / C2 ETHANE / C3 PROPANE /
17   CO2 CARBON-DIOXIDE / N2 NITROGEN / H2O WATER / H2S HYDROGEN-SULFIDE
18
19 PROPERTIES PENG-ROB
20
21 STREAMS
22   RAW_GAS TEMP=25 PRES=5 MOLE-FLOW=1000
23   COMP C1=0.92 C2=0.045 C3=0.02 N2=0.01 CO2=0.005 H2O=0.001
24
25 BLOCK INLET_SEP FLASH2
26   PARAM TEMP=25 PRES=5
27
28 BLOCK DEHYDRATOR SEP
29   SPEC FRAC H2O=0.999 IN WATER_PURGE
30   SPEC MAX H2O IN DRY_GAS = 16 MG/M3
31
32 BLOCK COMP1 COMPR
33   PARAM PRES=20 EFF=0.70
34
35 BLOCK IC1 HEATER
36   PARAM TEMP=40 PRES=19
37
38 BLOCK COMP2 COMPR
39   PARAM PRES=63 EFF=0.70
40
41 BLOCK IC2 HEATER
42   PARAM TEMP=40 PRES=60
43
44 BLOCK COMP3 COMPR
45   PARAM PRES=200 EFF=0.70
46
47 BLOCK AFTERCOOLER HEATER
48   PARAM TEMP=40 PRES=195
49
50 BLOCK STORAGE TANK
51   PARAM PRES=200 TEMP=40
52
53 BLOCK DISPENSER VALVE
54   PARAM PRES=250 TO 300
55
56 RESULTS
57   REPORT STREAMS DUTIES COMPRESSION-WORK GAS-QUALITY

```

Listing 18.3: CNG_Compression_BaseCase.apw (complete)

```

1  TITLE 'Green Hydrogen Electrolysis - Base Case'
2
3  FLOWSHEET
4  BLOCK WATER_PUMP IN = DI_WATER
5  BLOCK FEED_HEATER IN = PUMPED_WATER
6  BLOCK ELECTROLYZER IN = HOT_WATER POWER
7  BLOCK GAS_SEPARATOR IN = STACK_OUT
8  BLOCK H2_DRYER IN = RAW_H2
9  BLOCK H2_COMPRESSOR IN = DRY_H2
10 BLOCK H2_STORAGE IN = COMPRESSED_H2
11
12 COMPONENTS
13   H2 HYDROGEN / O2 OXYGEN / H2O WATER / N2 NITROGEN
14
15 PROPERTIES STEAMNBS
16
17 STREAMS
18   DI_WATER TEMP=25 PRES=1 MASS-FLOW=9000 COMP H2O=1

```



```

19  POWER WORK=50 KWH/KG-H2
20
21  BLOCK WATER_PUMP PUMP
22    PARAM PRES=30 EFF=0.75
23
24  BLOCK FEED_HEATER HEATER
25    PARAM TEMP=80 PRES=30
26    CASE PEM TEMP=80
27    CASE ALKALINE TEMP=90
28    CASE SOEC TEMP=750
29
30  BLOCK ELECTROLYZER RSTOIC
31    PARAM PRES=30
32    REAC WATER_SPLIT H2O -> H2 + 0.5 O2 CONV=0.90
33    SPEC EFFICIENCY PEM=0.68 ALKALINE=0.65 SOEC=0.82
34
35  BLOCK GAS_SEPARATOR SEP
36    SPEC FRAC H2=0.999 IN RAW_H2
37    SPEC FRAC O2=0.999 IN O2_BYPRODUCT
38    SPEC FRAC H2O=0.999 IN WATER_RECYCLE
39
40  BLOCK H2_DRYER SEP
41    SPEC FRAC H2O=0.999 IN WATER_RECYCLE
42
43  BLOCK H2_COMPRESSOR COMPR
44    PARAM PRES=350 EFF=0.72
45
46  BLOCK H2_STORAGE TANK
47    PARAM PRES=350 TEMP=40
48
49  RESULTS
50    REPORT H2-PRODUCTION POWER WATER-USE COMPRESSION-WORK

```

Listing 18.4: Electrolysis_BaseCase.apw (complete)

```

1  TITLE 'Methanol Synthesis from Syngas - Base Case'
2
3  FLOWSHEET
4    BLOCK MIXER IN = SYNGAS RECYCLE_GAS
5    BLOCK METHANOL_REACTOR IN = MIXED_FEED
6    BLOCK CONDENSER IN = REACTOR_OUT
7    BLOCK FLASH IN = CONDENSER_OUT
8    BLOCK RECYCLE_SPLIT IN = CONDENSER_VAPOR
9    BLOCK RECYCLE_COMP IN = RECYCLE_GAS_RAW
10   BLOCK DISTILLATION IN = CRUDE_METHANOL
11
12  COMPONENTS
13    H2 HYDROGEN / CO CARBON-MONOXIDE / CO2 CARBON-DIOXIDE /
14    CH4 METHANE / N2 NITROGEN / METHANOL METHANOL / H2O WATER
15
16  PROPERTIES PENG-ROB
17    FREE-WATER STEAMNBS
18
19  STREAMS
20    SYNGAS TEMP=220 PRES=70 MOLE-FLOW=10000
21    COMP H2=0.65 CO=0.25 CO2=0.05 CH4=0.03 N2=0.02
22
23  BLOCK MIXER MIXER
24    PARAM PRES=70
25
26  BLOCK METHANOL_REACTOR RSTOIC
27    PARAM TEMP=250 PRES=70
28    REAC MEOH_CO CO + 2 H2 -> METHANOL CONV=0.40
29    REAC MEOH_CO2 CO2 + 3 H2 -> METHANOL + H2O CONV=0.20
30
31  BLOCK CONDENSER HEATER
32    PARAM TEMP=40 PRES=68
33
34  BLOCK FLASH FLASH2
35    PARAM TEMP=40 PRES=68
36
37  BLOCK RECYCLE_SPLIT FSPLIT

```

```
38  PARAM PURGE-FRACTION=0.05
39
40  BLOCK RECYCLE_COMP COMPR
41  PARAM PRES=70 EFF=0.72
42
43  BLOCK DISTILLATION RADFRAC
44  PARAM NSTAGE=20 FEED-STAGE=10 REFLUX-RATIO=1.5
45  SPEC DISTILLATE METHANOL=0.995
46
47  RESULTS
48  REPORT CONVERSION RECYCLE PURGE METHANOL-PRODUCT DUTIES
```

Listing 18.5: Methanol_Synthesis_BaseCase.apw (complete)

Chapter 19

Python Code for All Figures

```
1 import matplotlib.pyplot as plt
2 import numpy as np
3
4 years = [1990,1995,2000,2005,2010,2015,2020,2025,2030,2035,2040,2045,2050]
5 fossil = [7500,7900,8500,9200,10500,11200,11500,11800,12000,11900,11700,10800,10000]
6 nuclear = [500,600,700,720,700,680,600,600,600,620,650,720,800]
7 renewables = [800,850,1000,1150,1500,2000,2500,3000,4000,5200,6000,7200,8000]
8 hydrogen = [0,0,0,0,0,0,50,200,450,800,1300,2000]
9
10 fig, ax = plt.subplots(figsize=(14, 8))
11 ax.stackplot(years, fossil, nuclear, renewables, hydrogen,
12             labels=['Fossil Fuels', 'Nuclear', 'Renewables', 'Hydrogen'],
13             colors=['#8B4513', '#FFA500', '#2E8B57', '#00BFFF'], alpha=0.85)
14 ax.set_xlabel('Year', fontsize=12)
15 ax.set_ylabel('Energy Consumption (Mtoe)', fontsize=12)
16 ax.set_title('Global Primary Energy Consumption (1990-2050)')
17 ax.legend(loc='upper left')
18 ax.grid(True, alpha=0.3)
19 ax.set_xlim(1990, 2050); ax.set_ylim(0, 22000)
20 plt.tight_layout()
21 plt.savefig('Figure_1_1_Energy_Consumption.png', dpi=300)
22 plt.show()
```

Listing 19.1: Energy consumption plotting script

```
1 import matplotlib.pyplot as plt
2 import numpy as np
3
4 length = np.linspace(0, 100, 200)
5 temp = 600 + 250 * (1 - np.exp(-length / 25))
6
7 fig, ax = plt.subplots(figsize=(12, 7))
8 ax.fill_between(length, 0, temp, alpha=0.3, color='red')
9 ax.plot(length, temp, 'r-', linewidth=3, label='Reactor Temperature')
10 ax.axvspan(0, 20, alpha=0.25, color='blue', label='Preheat Zone')
11 ax.axvspan(20, 70, alpha=0.25, color='yellow', label='Reaction Zone')
12 ax.axvspan(70,100, alpha=0.25, color='orange', label='Equilibrium Zone')
13 ax.set_xlabel('Reactor Tube Length (%)')
14 ax.set_ylabel('Temperature (degC)')
15 ax.set_title('Temperature Profile Along SMR Reactor Tube')
16 ax.legend(); ax.grid(True, alpha=0.3)
17 ax.set_xlim(0,100); ax.set_ylim(0,950)
18 plt.tight_layout()
19 plt.savefig('Figure_3_2_SMR_Temperature_Profile.png', dpi=300)
20 plt.show()
```

Listing 19.2: SMR temperature profile plotting script

```
1 import matplotlib.pyplot as plt
2 import matplotlib.patches as patches
3
4 fig, ax = plt.subplots(figsize=(10, 8))
5 ax.set_xlim(0, 10); ax.set_ylim(0, 8)
6 ax.axis('off')
7
8 # Boxes
9 boxes = [
10     ('Goal and Scope Definition', 2, 6.5),
11     ('Life Cycle Inventory (LCI)', 2, 5.0),
12     ('Life Cycle Impact Assessment (LCIA)', 2, 3.5),
```

```
13     ('Interpretation', 2, 2.0)
14 ]
15
16 for text, x, y in boxes:
17     rect = patches.Rectangle((x, y-0.5), 6, 0.8, linewidth=2, edgecolor='blue', facecolor
18     = 'lightblue')
19     ax.add_patch(rect)
20     ax.text(x+3, y, text, ha='center', va='center', fontsize=12)
21
22 # Arrows
23 ax.arrow(5, 5.7, 0, -0.8, head_width=0.3, head_length=0.2, fc='black')
24 ax.arrow(5, 4.2, 0, -0.8, head_width=0.3, head_length=0.2, fc='black')
25 ax.arrow(5, 2.7, 0, -0.8, head_width=0.3, head_length=0.2, fc='black')
26
27 # Feedback loop
28 ax.plot([5, 5, 2], [1.5, 1, 1], color='black')
29 ax.arrow(2, 1, 0, 1.2, head_width=0.3, head_length=0.2, fc='black')
30
31 plt.title('ISO 14040/14044 LCA Framework', fontsize=14, fontweight='bold')
32 plt.tight_layout()
33 plt.savefig('Figure_10_1_LCA_Flowchart.png', dpi=300)
34 plt.show()
```

Listing 19.3: LCA flowchart plotting script

Chapter 20

Data Tables and Appendices

20.1 Complete Stream Table

Table 20.1: Complete Stream Table for SMR Plant

Stream	T (°C)	P (bar)	Flow (kg/hr)	H ₂ %	CO%	CO ₂ %	CH ₄ %
NG_FEED	25	20	5,000	0	0	0.5	92
STEAM	250	25	9,000	0	0	0	0
REFORMER_OUT	850	19	14,000	40	12	8	10
HTS_OUT	400	18.5	14,000	48	3	17	10
LTS_OUT	210	18	14,000	50	0.5	20	10
H2_PRODUCT	40	25	1,000	99.99	0	0	0.01
CO2_CAPTURE	40	1.5	13,200	0	0	95	0

20.2 Economic Parameters Database

Table 20.2: Economic Parameters

Parameter	Value	Unit
Discount rate (nominal)	7	%
Plant life (SMR)	25	years
Plant life (electrolysis)	20	years
NG price (US)	3.50–5.50	\$/MMBtu
NG price (EU)	8.00–12.00	\$/MMBtu
Electricity price (US)	0.07	\$/kWh
Electricity price (EU)	0.12–0.20	\$/kWh
Electricity price (China)	0.05	\$/kWh
Carbon tax (EU ETS)	65–100	\$/tonne
45Q tax credit (US)	85	\$/tonne
Carbon price (UK)	50	\$/tonne
H ₂ price (grey)	1.50–2.50	\$/kg
H ₂ price (green)	3.00–6.00	\$/kg

20.3 Physical Properties of Fuels

20.4 Technology Readiness Level Definitions

Table 20.3: Physical Properties

Property	H₂	CH₄	CH₃OH	Diesel
Molecular weight (g/mol)	2.016	16.04	32.04	~170
LHV (MJ/kg)	120.0	50.0	20.0	42.7
Boiling point (°C)	-252.9	-161.5	64.7	170–360
Flammability limits (vol%)	4–75	5–15	6.7–36	0.6–7.5
MIE (mJ)	0.018	0.29	0.14	0.24
Autoignition temp (°C)	520	600	385	210

Table 20.4: Technology Readiness Levels (TRL)

TRL	Description	Example
1	Basic principles observed	Microwave plasma SMR
2	Concept formulated	AEM electrolysis (early)
3	Proof of concept	e-Methanol (DAC)
4	Component validation in lab	AEM electrolysis
5	Component validation in environment	e-SMR
6	System prototype demonstration	BeBOP methanol
7	System pilot demonstration	SOEC electrolysis
8	System commissioned	Blue H ₂ (first plants)
9	Commercial operation	CNG, SMR grey, Alkaline

Chapter 21

Glossary of Terms

Term	Definition
Alternative fuel	Any fuel other than conventional fossil fuels (gasoline, diesel)
Aspen Plus	Process simulation software used for chemical process modeling
Blue hydrogen	Hydrogen produced from fossil fuels with carbon capture and storage
Capacity factor	Ratio of actual output to maximum possible output over a period
Carbon capture	Process of capturing CO ₂ from industrial emissions
Compressed natural gas	Natural gas compressed to 200–250 bar for vehicle fuel
Cradle-to-grave	Life cycle assessment covering all stages from extraction to disposal
Electrolyzer	Device that splits water into hydrogen and oxygen using electricity
Engineered fuel	Energy carrier produced through chemical/electrochemical conversion
Green hydrogen	Hydrogen produced from renewable energy via electrolysis
Levelized cost	Average cost of producing energy over plant lifetime
Life cycle assessment	Method to assess environmental impacts across product lifetime
Methanol	CH ₃ OH, liquid fuel and chemical feedstock
Multi-criteria decision analysis	Decision-making method evaluating multiple criteria simultaneously
PEM electrolysis	Proton exchange membrane electrolysis for hydrogen production
Steam methane reforming	Main industrial method for hydrogen production from natural gas
Techno-economic analysis	Study of technical performance and economic viability
Technology readiness level	Scale measuring technology maturity from 1 to 9
Water-gas shift	Reaction converting CO and H ₂ O to CO ₂ and H ₂

Chapter 22

Complete Nomenclature Table

Symbol	Description	Units
A	Pre-exponential factor	varies
a	Peng-Robinson energy parameter	$\text{J}\cdot\text{m}^3/\text{mol}^2$
b	Peng-Robinson volume parameter	m^3/mol
CF	Capacity factor	dimensionless
CRF	Capital recovery factor	dimensionless
E	Energy	J
E_a	Activation energy	J/mol
Ex	Exergy	J/kg
F	Faraday constant	C/mol
G	Gibbs free energy	J
h	Specific enthalpy	J/kg
H	Enthalpy	J
I	Current	A
j	Current density	A/m^2
k	Reaction rate constant	varies
K	Equilibrium constant	varies
$\dot{m}$	Mass flow rate	kg/s
n	Number of electrons	dimensionless
P	Pressure	Pa
$\dot{Q}$	Heat transfer rate	W
r	Reaction rate	$\text{mol}/\text{m}^3\cdot\text{s}$
R	Universal gas constant	$\text{J}/\text{mol}\cdot\text{K}$
S	Entropy	J/K
T	Temperature	K
U	Internal energy	J
V	Volume	m^3
$\dot{W}$	Work rate	W
α	Charge transfer coefficient	dimensionless
η	Overpotential	V
η	Efficiency	dimensionless
κ	Peng-Robinson parameter	dimensionless
ω	Acentric factor	dimensionless

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